

Trait-based Ecology III

Today's Agenda:

- Quiz
- Traits and ecosystem function

Syllabus update

- Removed one “TBD” assignment and rolled the points into the Project 4
 - Project 4 will allow you to recycle a lot of text from previous projects
- Remaining Assignment will be a group project
- We'll go over these two items next Monday

Three things remaining:

- 1) Project 3 (Due Friday)
- 2) Group project (due due April 30th)
- 3) Project 4 (due May 9th, unless there are USF restrictions I don't know about)

Why focus on traits?

- Generality:
 - Rather than focusing on particular species in particular systems, we can say things about all species in all systems (or at least a larger subset of species/systems).
- Quantitative:
 - We move from species (categorical) to continuous measurements
- Links to mechanism
 - We can focus on traits that are linked to particular mechanisms
 - E.g., if we want to work on predation, mouth size might be useful.
- More predictive
 - Allow us to say something about potential future scenarios

Why focus on traits?

- Generality:
 - Rather than focusing on particular species in particular systems, we can say things about all species in all systems (or at least a larger subset of species/systems).
- Quantitative:
 - We move from species (categorical) to continuous measurements
- Links to mechanism
 - We can focus on traits that are linked to particular mechanisms
 - E.g., if we want to work on predation, mouth size might be useful.
- More predictive
 - Allow us to say something about potential future scenarios

Traits and ecosystem function

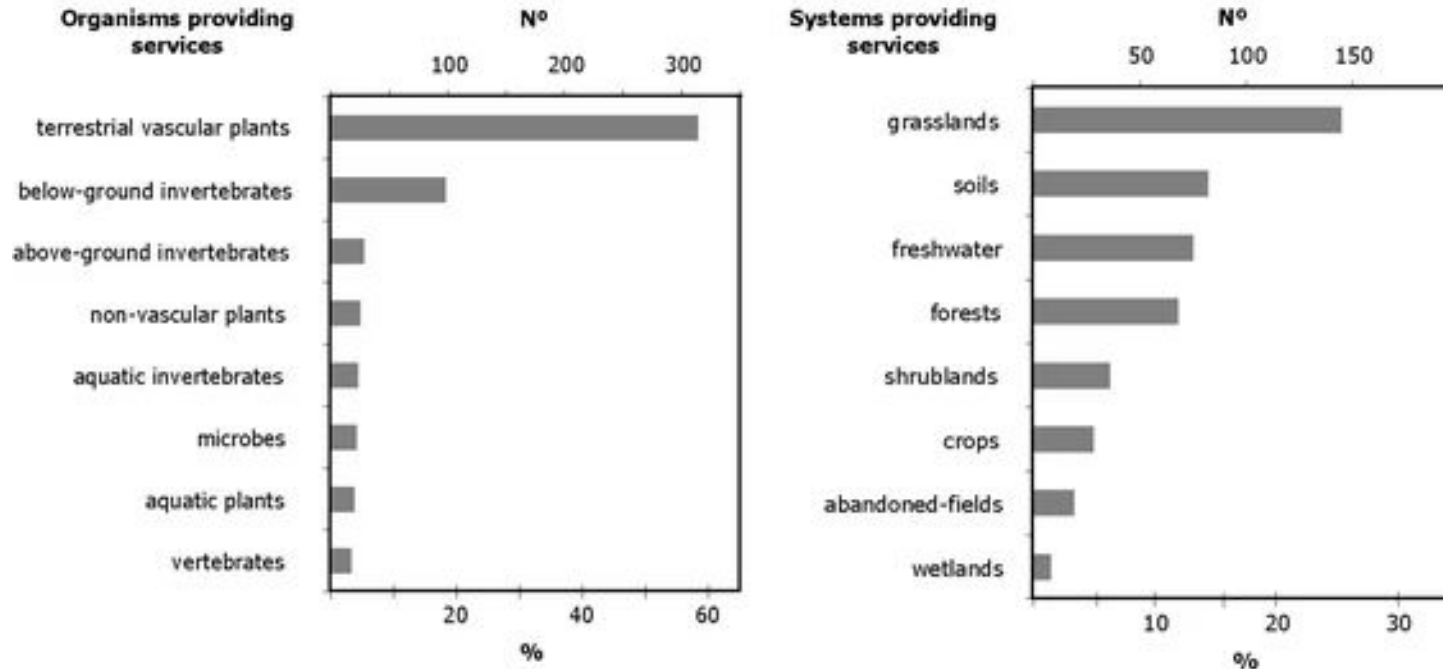
- Which organisms?
- Which traits map onto which services?
- How much do traits vs environment matter?

Towards an assessment of multiple ecosystem processes and services via functional traits

Francesco de Bello · Sandra Lavorel · Sandra Díaz · Richard Harrington · Johannes H. C. Cornelissen · Richard D. Bardgett · Matty P. Berg · Pablo Cipriotti · Christian K. Feld · Daniel Hering · Pedro Martins da Silva · Simon G. Potts · Leonard Sandin · Jose Paulo Sousa · Jonathan Storkey · David A. Wardle · Paula A. Harrison

What are people studying?

RANKING OF ORGANISMS AND SYSTEMS PROVIDING SERVICES AS FOUND IN THE LITERATURE

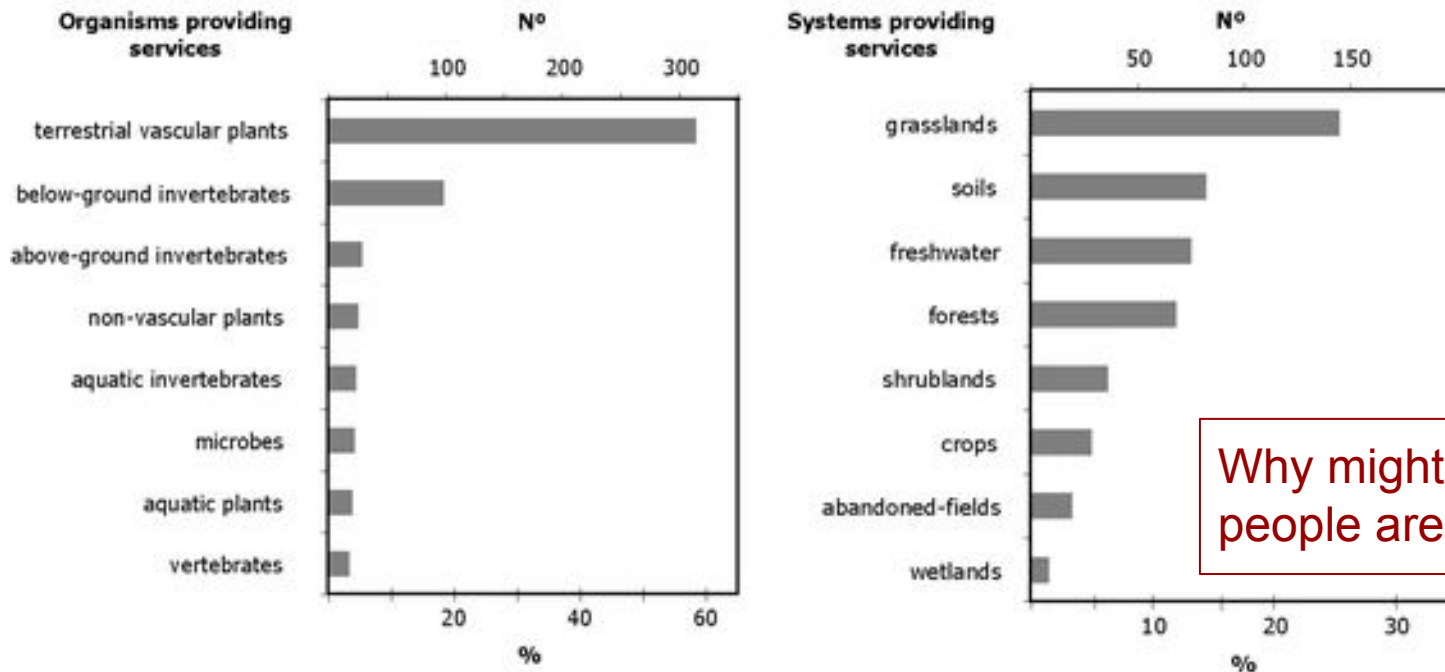


Towards an assessment of multiple ecosystem processes and services via functional traits

Francesco de Bello · Sandra Lavorel · Sandra Díaz · Richard Harrington · Johannes H. C. Cornelissen · Richard D. Bardgett · Matty P. Berg · Pablo Cipriotti · Christian K. Feld · Daniel Hering · Pedro Martins da Silva · Simon G. Potts · Leonard Sandin · Jose Paulo Sousa · Jonathan Storkey · David A. Wardle · Paula A. Harrison

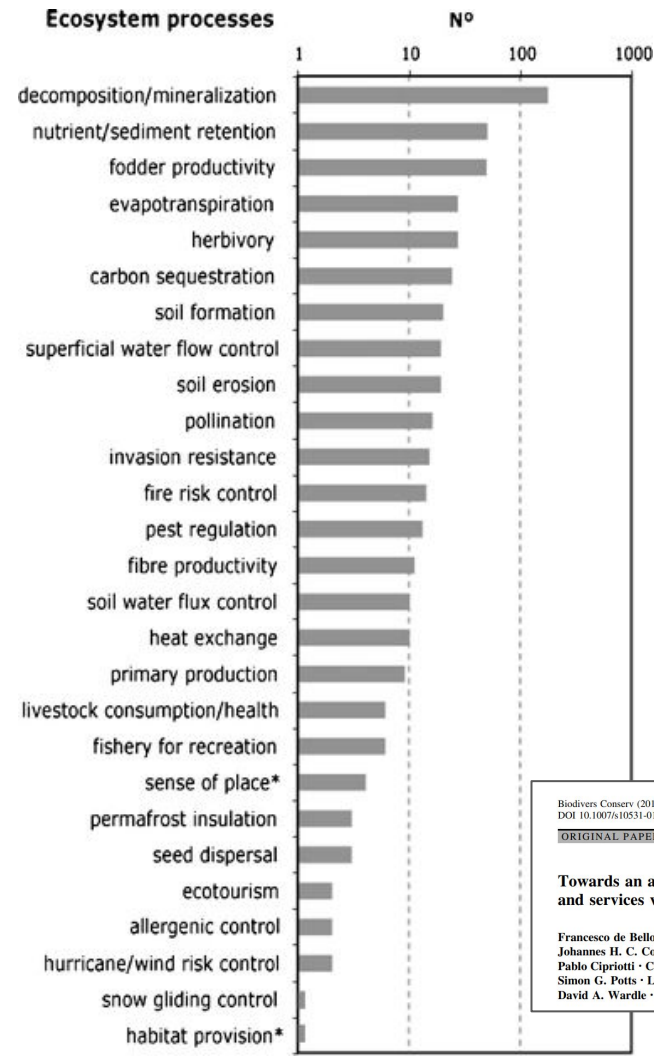
What are people studying?

RANKING OF ORGANISMS AND SYSTEMS PROVIDING SERVICES AS FOUND IN THE LITERATURE



Why might these be what people are studying?

What are people studying?



Biodivers. Conserv. (2010) 19:2873–2893
DOI 10.1007/s10531-010-9850-9

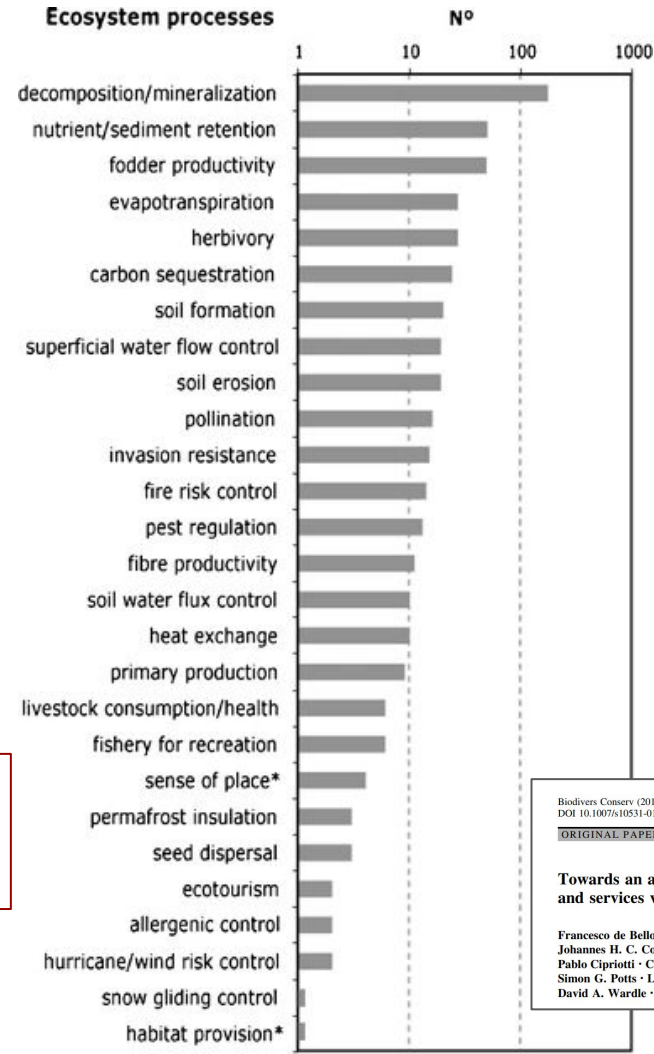
ORIGINAL PAPER

Towards an assessment of multiple ecosystem processes and services via functional traits

Francesco de Bello · Sandra Lavorel · Sandra Díaz · Richard Harrington · Johannes H. C. Cornelissen · Richard D. Bardgett · Matty P. Berg · Pablo Cipriotti · Christian K. Feld · Daniel Hering · Pedro Martins da Silva · Simon G. Potts · Leonard Sandin · Jose Paulo Sousa · Jonathan Storkey · David A. Wardle · Paula A. Harrison

What are people studying?

Why might these be what people are studying?



Biodivers. Conserv. (2010) 19:2873–2893
DOI 10.1007/s10531-010-9850-9

ORIGINAL PAPER

Towards an assessment of multiple ecosystem processes and services via functional traits

Francesco de Bello · Sandra Lavorel · Sandra Díaz · Richard Harrington · Johannes H. C. Cornelissen · Richard D. Bardgett · Matty P. Berg · Pablo Cipriotti · Christian K. Feld · Daniel Hering · Pedro Martins da Silva · Simon G. Potts · Leonard Sandin · Jose Paulo Sousa · Jonathan Storkey · David A. Wardle · Paula A. Harrison

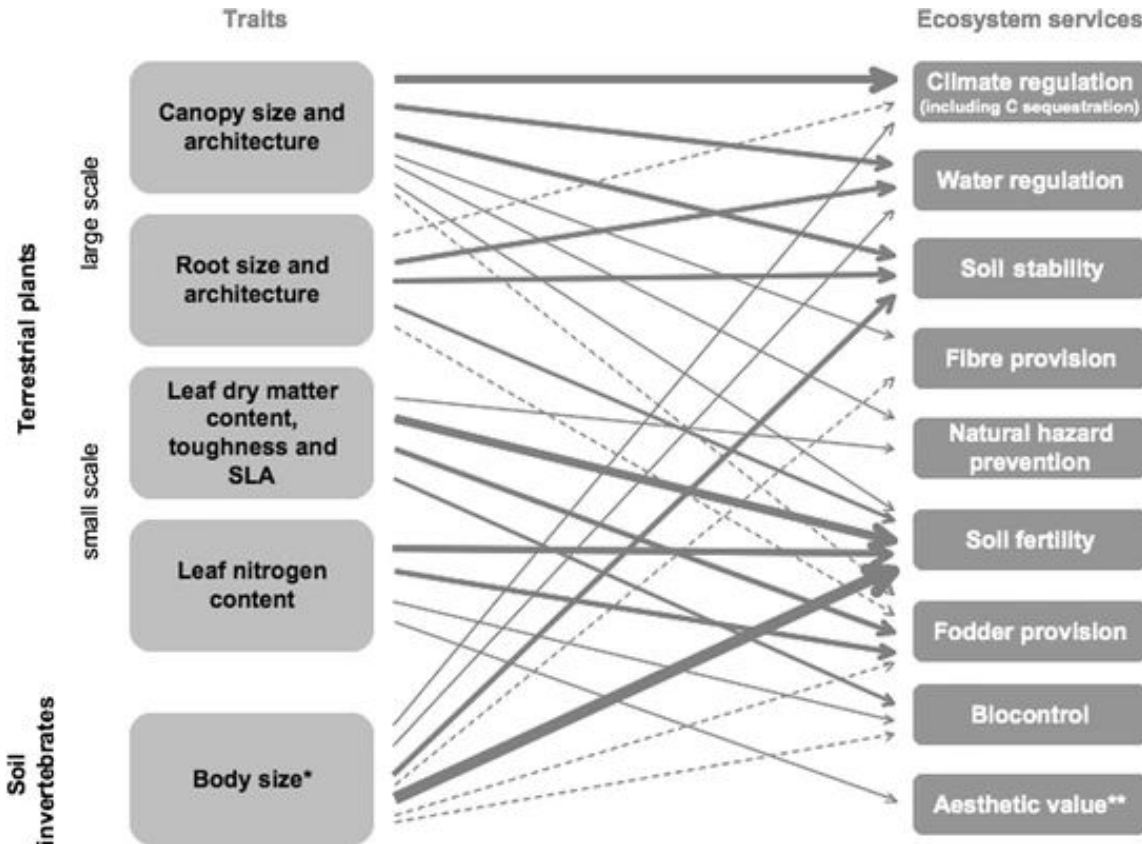
Towards an assessment of multiple ecosystem processes and services via functional traits

Francesco de Bello · Sandra Lavorel · Sandra Díaz · Richard Harrington · Johannes H. C. Cornelissen · Richard D. Bardgett · Matty P. Berg · Pablo Cipriotti · Christian K. Feld · Daniel Hering · Pedro Martins da Silva · Simon G. Potts · Leonard Sandin · Jose Paulo Sousa · Jonathan Storkey · David A. Wardle · Paula A. Harrison

What are people studying?

Traits

Ecosystem services



Ecosystem function axes

- Lots of ecosystem functions we might study
- Some correlated
- Wanted to understand major variation

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

Published online: 22 September 2021

Open access

 Check for updates

Mirco Migliavacca^{1,2,8,9,12}, Talie Musavi¹, Miguel D. Mahecha^{12,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,10}, Dennis D. Baldocchi⁸, Oscar Perez-Priego¹, Rune Christiansen¹, Jonas Peters⁴, Karen Anderson⁷, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹¹, Damien Bone¹², Nina Buchmann¹, Silvia Caldararu¹, Arnaud Carrara¹, Nuno Carvalhais¹⁰, Alessandro Cecatti¹², Jiquan Chen¹¹, Jamie Cleverly^{10,11}, Edoardo Cremonesi¹², Ankur R. Desai¹², Tarek S. El-Madany¹², Martha M. Farella¹², Marcos Fernández-Martínez²⁴, Gianluca Filippa¹², Matthias Forkel¹², Marta Galvagno¹², Ulisse Gomasasca¹, Christopher M. Gough¹⁰, Mathias Göckede¹, Andreas Ibrom¹², Hiroki Iwata¹⁰, Ivan A. Janssens¹⁰, Martin Jung¹, Jens Kattge¹², Trevor F. Keenan^{10,11}, Alexander Knohl^{10,11}, Hideki Kobayashi¹², Guido Krömer^{10,11}, Beverly E. Law¹⁰, Michael J. Liddell¹², Xuanlong Ma¹⁰, Ivan Mammarella¹², David Martin¹, Craig Macfarlane¹⁰, Giorgio Matteucci¹⁰, Leonardo Montagnani^{10,11}, Daniel E. Pabon-Moreno¹, Cinzia Panigada¹², Dario Papale¹², Elise Pendall¹⁴, Josep Penuelas^{10,11}, Richard P. Phillips¹², Peter B. Reich^{14,10,11}, Micol Rossini¹², Eyal Rotenberg¹⁰, Russell L. Scott¹¹, Clement Stahl¹², Ulrich Weber¹, Georg Wohlfahrt¹⁰, Sebastian Wolf¹⁰, Ian J. Wright^{14,11}, Dan Yaki¹⁰, Sönke Zaehe¹ & Markus Reichstein^{1,2,3,4,12}

Ecosystem function axes

Ecosystem services measured:

- WUEt, variant of water-use efficiency, but based on transpiration estimates
- underlying water-use efficiency (uWUE)
- maximum and mean basal ecosystem respiration (Rbmax and Rb, respectively)
- maximum net ecosystem productivity (NEPmax),
- maximum dry canopy surface conductance (Gsmax)
- maximum gross CO₂ uptake at light saturation (GPPsat)

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

Published online: 22 September 2021

Open access

 Check for updates

Mirco Migliavacca^{1,2,8,9,10}, Tatle Musavi¹, Miguel D. Mahecha^{12,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,10}, Dennis D. Baldocchi⁸, Oscar Perez-Priego¹, Rune Christiansen⁹, Jonas Peters⁹, Karen Anderson⁹, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹⁰, Damien Bonal¹², Nina Buchmann¹, Silvia Caldararu¹, Arnaud Carrara⁹, Nuno Carvalhais¹⁰, Alessandro Cescatti¹, Jiquan Chen¹, Jamie Cleverly^{10,11}, Edoardo Cremonese¹, Ankur R. Desai¹², Tarek S. El-Madany¹, Martha M. Farello¹², Marcos Fernández-Martínez²⁴, Gianluca Filippa²⁵, Matthias Forkel²⁵, Marta Galvagno²⁵, Ulisse Gomasasca¹, Christopher M. Gough²⁶, Mathias Göckede¹, Andreas Ibrom²⁷, Hiroki Ikawa²⁸, Ivan A. Janssens²⁹, Martin Jung¹, Jens Kattge¹, Trevor F. Keenan^{30,31}, Alexander Knohl^{32,33}, Hideki Kobayashi³², Guido Krömer^{33,34}, Beverly E. Law³⁵, Michael J. Liddell³⁶, Xuanlong Ma³⁶, Ivan Mammarella³⁷, David Martin¹, Craig Macfarlane³⁸, Giorgio Matteucci³⁹, Leonardo Montagnani^{40,41}, Daniel E. Pabon-Moreno¹, Cinzia Panigada⁴², Dario Papale⁴³, Elise Pendall⁴⁴, Josep Penuelas^{45,46}, Richard P. Phillips⁴⁷, Peter B. Reich^{44,46,48}, Micol Rossini⁴⁹, Eyal Rotenberg⁵⁰, Russell L. Scott⁵¹, Clement Stahl⁵², Ulrich Weber¹, Georg Wohlfahrt⁵³, Sebastian Wolf⁵⁴, Ian J. Wright^{44,55}, Dan Yaki⁵⁶, Sönke Zaehle¹ & Markus Reichstein^{1,2,34,47}

Ecosystem function axes

Continued:

- stomatal slope at ecosystem scale (G1)
- maximum evapotranspiration (ET_{max}),
- EF amplitude (EF_{ampl})
- evaporative fraction (EF) (that is, the ratio between latent heat flux and available energy, indicative of energy partitioning)
- apparent carbon-use efficiency (aCUE) (that is, the remaining fraction of carbon entering the ecosystem).

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

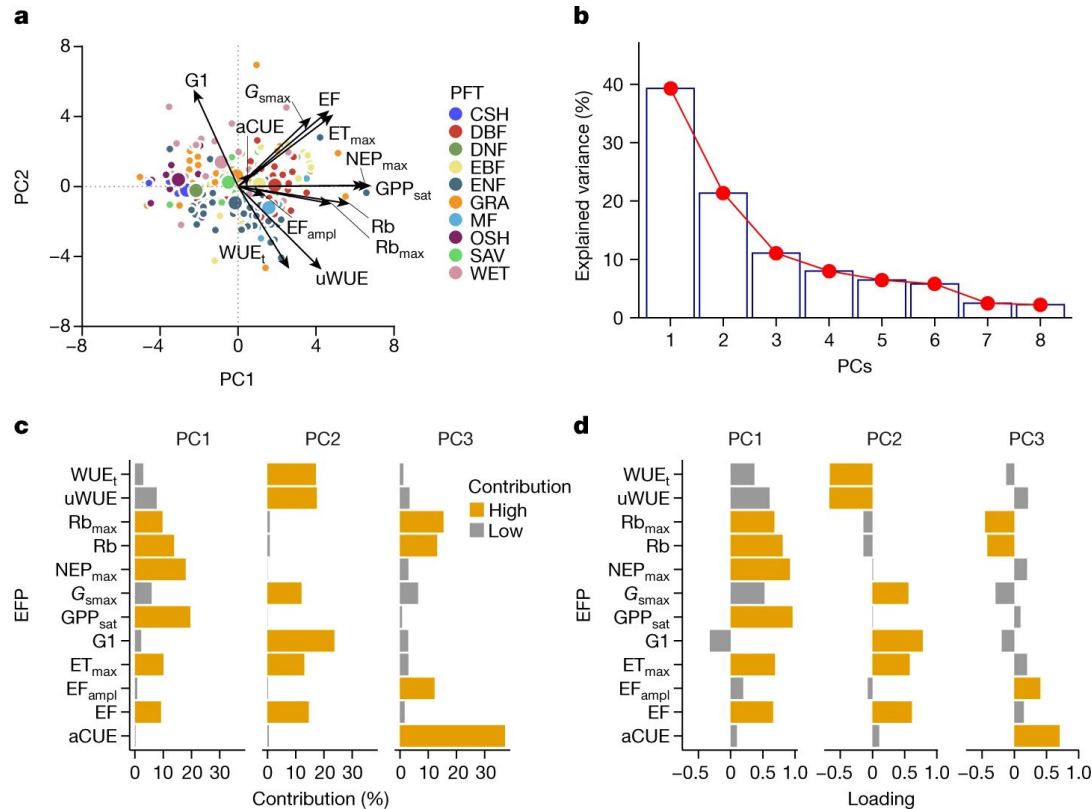
Published online: 22 September 2021

Open access

 Check for updates

Mirco Migliavacca^{1,2,8,9,10}, Tatle Musavi¹, Miguel D. Mahecha^{12,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,10}, Dennis D. Baldocchi⁵, Oscar Perez-Priego¹, Rune Christiansen⁶, Jonas Peters⁷, Karen Anderson⁷, Michael Bahn¹⁰, T. Andrew Black¹, Peter D. Blanken¹⁰, Damien Bone¹¹, Nina Buchmann¹, Silvia Caldararu¹, Arnaud Carrara¹, Nuno Carvalhais¹⁰, Alessandro Cecatti¹¹, Jiquan Chen¹, Jamie Cleverly^{10,11}, Edoardo Cremonese¹, Ankur R. Desai¹², Tarek S. El-Madany¹, Martha M. Farella¹², Marcos Fernández-Martínez²⁴, Gianluca Filippa²⁵, Matthias Forkel²⁵, Marta Galvagno²⁵, Ulisse Gomasca¹, Christopher M. Gough²⁶, Mathias Göckede¹, Andreas Ibrom²⁷, Hiroki Ikawa²⁸, Ivan A. Janssens²⁹, Martin Jung¹, Jens Kattge¹, Trevor F. Keenan^{30,31}, Alexander Knohl^{32,33}, Hideki Kobayashi³³, Guido Kraemer³³, Beverly E. Law³⁴, Michael J. Liddell³⁵, Xuanlong Ma³⁶, Ivan Mammarella³⁷, David Martin¹, Craig Macfarlane³⁸, Giorgio Matteucci³⁹, Leonardo Montagnani^{40,41}, Daniel E. Pabon-Moreno¹, Cinzia Panigada⁴², Dario Papale⁴³, Elise Pendall⁴⁴, Josep Penuelas^{45,46}, Richard P. Phillips⁴⁷, Peter B. Reich^{44,46,48}, Micol Rossini⁴⁹, Eyal Rotenberg⁵⁰, Russell L. Scott⁵¹, Clement Stahl⁵², Ulrich Weber¹, Georg Wohlfahrt⁵³, Sebastian Wolf⁵⁴, Ian J. Wright^{44,55}, Dan Yaki⁵⁶, Sönke Zaehle¹ & Markus Reichstein^{1,2,34,47}

Ecosystem function axes



Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

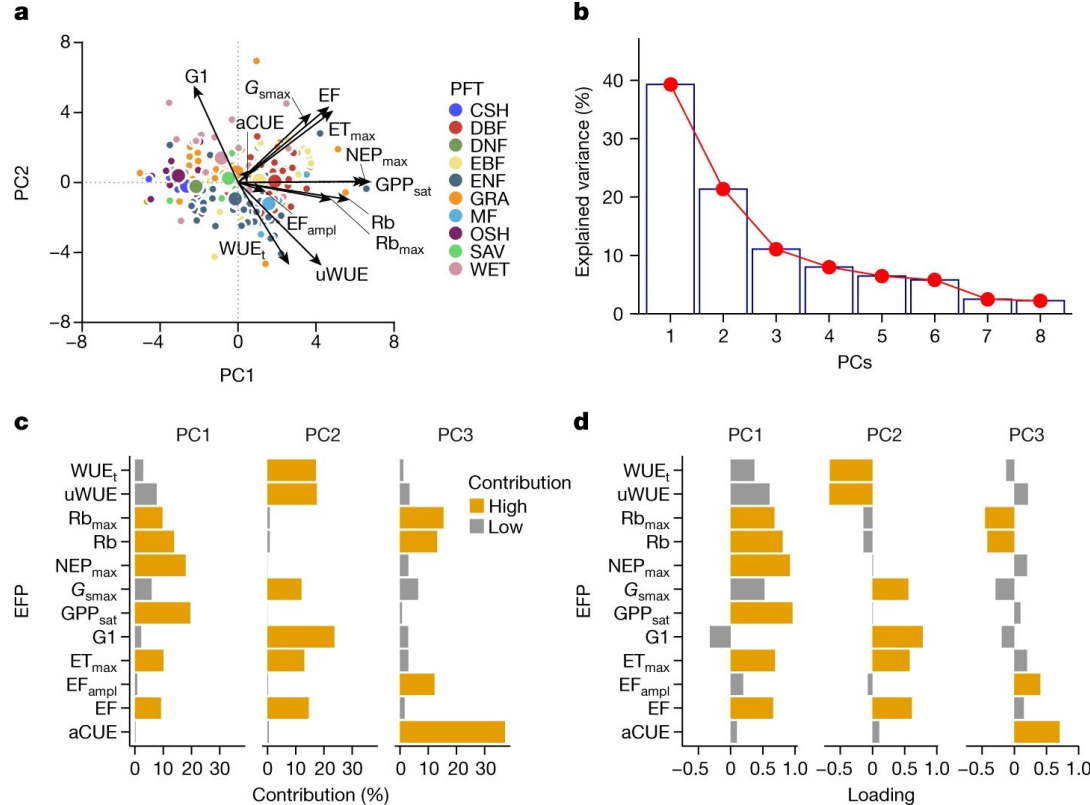
Published online: 22 September 2021

Open access

[Check for updates](#)

Mirco Migliavacca^{1,2,8,9,10}, Tatle Musavi¹, Miguel D. Mahecha^{12,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,10}, Dennis D. Baldocchi⁵, Oscar Perez-Priego¹, Rune Christiansen⁶, Jonas Peters⁷, Karen Anderson⁷, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹², Damien Bone¹³, Nina Buchmann¹⁴, Silvia Caldararu¹⁵, Arnaud Carrara¹⁶, Nuno Carvalhais¹⁷, Alessandro Ceccati¹⁸, Jiquan Chen¹⁹, Jamie Cleverly^{20,21}, Edoardo Cremonesi²², Ankur R. Desai²³, Tarek S. El-Madany²⁴, Martha M. Farello²⁵, Marcos Fernández-Martínez²⁶, Gianluca Filippa²⁷, Matthias Forkel²⁸, Marta Galvagno²⁹, Ulisse Gomasca³⁰, Christopher M. Gough³¹, Mathias Göckede³², Andreas Ibrom³³, Hiroki Iwasa³⁴, Ivan A. Janssens³⁵, Martin Jung³⁶, Jens Kattge³⁷, Trevor F. Keenan^{38,39}, Alexander Knohl^{40,41}, Hideki Kobayashi⁴², Guido Krömer^{43,44}, Beverly E. Law⁴⁵, Michael J. Liddell⁴⁶, Xuanlong Ma⁴⁷, Ivan Mammarella⁴⁸, David Martin⁴⁹, Craig Macfarlane⁵⁰, Giorgio Matteucci⁵¹, Leonardo Montagnani^{52,53}, Daniel E. Pabon-Moreno⁵⁴, Cinzia Panigada⁵⁵, Dario Papale⁵⁶, Elise Pendall⁵⁷, Josep Penuelas^{58,59}, Richard P. Phillips⁶⁰, Peter B. Reich^{61,62,63}, Micol Rossini⁶⁴, Eyal Rotenberg⁶⁵, Russell L. Scott⁶⁶, Clement Stahl⁶⁷, Ulrich Weber⁶⁸, Georg Wohlfahrt⁶⁹, Sebastian Wolf⁷⁰, Ian J. Wright^{44,61}, Dan Yakir⁷¹, Sonke Zaehle⁷² & Markus Reichstein^{1,2,3,4,73}

Ecosystem function axes



PC1 = maximum ecosystem productivity
 PC2 = ecosystem water-use strategies
 PC3 = ecosystem carbon-use efficiency

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

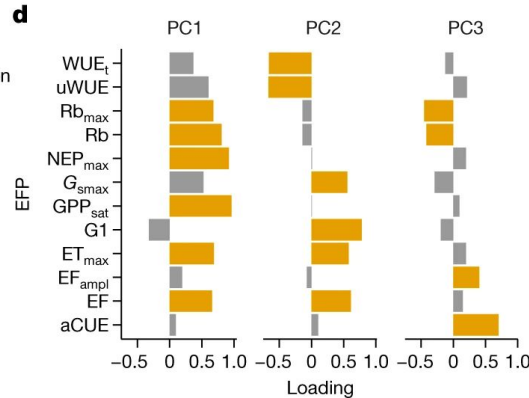
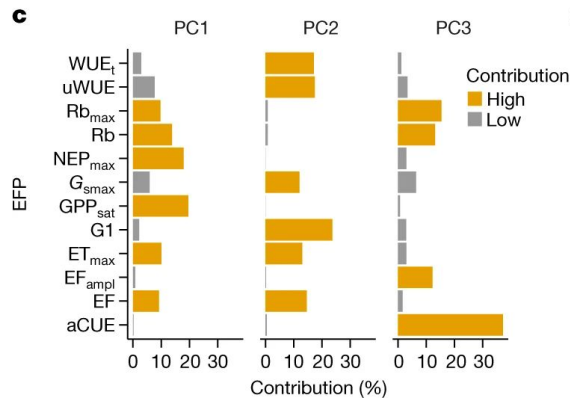
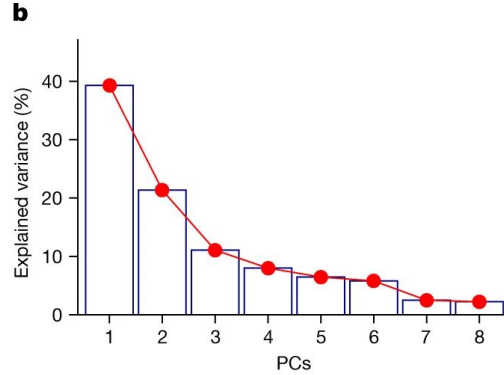
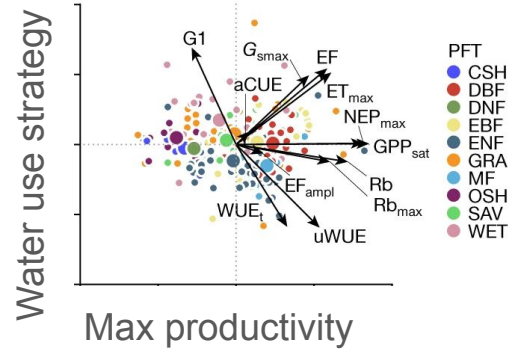
Published online: 22 September 2021

Open access

[Check for updates](#)

Mirco Migliavacca^{1,2,8,9,10}, Tatle Musavi¹, Miguel D. Mahecha^{12,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,10}, Dennis D. Baldocchi⁵, Oscar Perez-Priego¹, Rune Christiansen⁶, Jonas Peters⁷, Karen Anderson⁷, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹², Damien Bonel¹³, Nina Buchmann¹⁴, Silvia Caldararu¹, Arnaud Carrara¹⁵, Nuno Carvalhais¹⁶, Alessandro Cecatti¹⁷, Jiquan Chen¹⁸, Jamie Cleverly^{19,20}, Edoardo Cremonesi²¹, Ankur R. Desai²², Tarek S. El-Madany²³, Martha M. Farella²⁴, Marcos Fernández-Martínez²⁴, Gianluca Filippa²⁵, Matthias Forkel²⁶, Marta Galvagno²⁷, Ulisse Gomarasca²⁸, Christopher M. Gough²⁹, Mathias Göckede³⁰, Andreas Ibrom³¹, Hiroki Iwasa³², Ivan A. Janssens³³, Martin Jung³⁴, Jens Kattge³⁵, Trevor F. Keenan^{36,37}, Alexander Knohl^{38,39}, Hideki Kobayashi⁴⁰, Guido Krömer^{41,42}, Beverly E. Law⁴³, Michael J. Liddell⁴⁴, Xuanlong Ma⁴⁵, Ivan Mammarella⁴⁶, David Martin⁴⁷, Craig MacFarlane⁴⁸, Giorgio Matteucci⁴⁹, Leonardo Montagnani^{50,51}, Daniel E. Pabon-Moreno⁵², Cinzia Panigada⁵³, Dario Papale⁵⁴, Elise Pendall⁵⁵, Josep Penuelas^{56,57}, Richard P. Phillips⁵⁸, Peter B. Reich^{54,59,60}, Micol Rossini⁶¹, Eyal Rotenberg⁶², Russell L. Scott⁶³, Clement Stahl⁶⁴, Ulrich Weber⁶⁵, Georg Wohlfahrt⁶⁶, Sebastian Wolf⁶⁷, Ian J. Wright^{68,69}, Dan Yakir⁷⁰, Sonke Zaehle⁷¹ & Markus Reichstein^{1,2,8,9,10}

Ecosystem function axes



Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

Published online: 22 September 2021

Open access

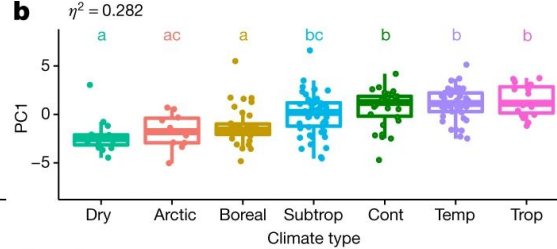
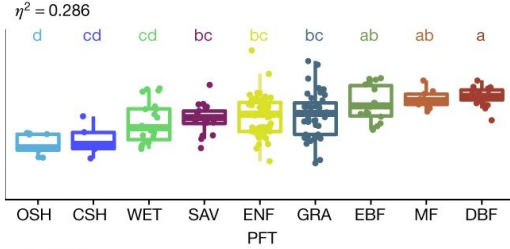
[Check for updates](#)

Mirco Migliavacca^{1,2,8,9,10}, Talie Musavi¹, Miguel D. Mahecha^{12,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,10}, Dennis D. Baldocchi⁵, Oscar Perez-Priego¹, Rune Christiansen⁶, Jonas Peters⁷, Karen Anderson⁷, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹⁰, Damien Bonal¹², Nina Buchmann¹³, Silvia Caldararu¹, Arnaud Carrara¹⁴, Nuno Carvalhais¹⁵, Alessandro Ceccati¹⁶, Jiquan Chen¹⁷, Jamie Cleverly^{18,19}, Edoardo Cremonesi²⁰, Ankur R. Desai²¹, Tarek S. El-Madany¹, Martha M. Farello²², Marcos Fernández-Martínez²⁴, Gianluca Filippa²⁵, Matthias Forkel²⁶, Marta Galvagno²⁷, Ulisse Gomarasca²⁸, Christopher M. Gough²⁹, Mathias Gökede³⁰, Andreas Ibrom³¹, Hiroki Iwasa³², Ivan A. Janssens³³, Martin Jung³⁴, Jens Kattge³⁵, Trevor F. Keenan^{36,37}, Alexander Knohl^{38,39}, Hideki Kobayashi⁴⁰, Guido Krömer^{41,42}, Beverly E. Law⁴³, Michael J. Liddell⁴⁴, Xuanlong Ma⁴⁵, Ivan Mammarella⁴⁶, David Martin⁴⁷, Craig MacFarlane⁴⁸, Giorgio Matteucci⁴⁹, Leonardo Montagnani^{50,51}, Daniel E. Pabon-Moreno⁵², Cinzia Panigada⁵³, Dario Papale⁵⁴, Elise Pendall⁵⁵, Josep Penuelas^{56,57}, Richard P. Phillips⁵⁸, Peter B. Reich^{59,60,61}, Micol Rossini⁶², Eyal Rotenberg⁶³, Russell L. Scott⁶⁴, Clement Stahl⁶⁵, Ulrich Weber⁶⁶, Georg Wohlfahrt⁶⁷, Sebastian Wolf⁶⁸, Ian J. Wright^{69,70}, Dan Yakir⁷¹, Sonke Zaehle⁷² & Markus Reichstein^{1,2,3,4,5,6,7,8,9,10}

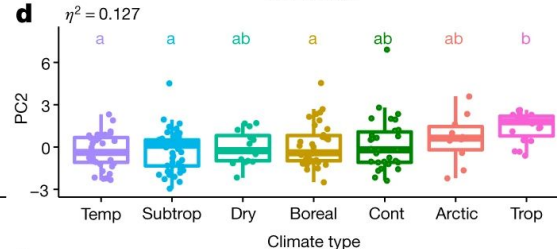
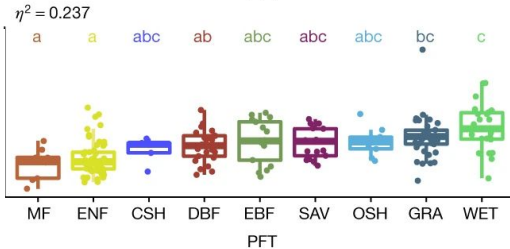
CSH = closed shrublands
DBF = deciduous broadleaved forest
DNF = deciduous needleleaf forests
EBF = evergreen broadleaved forest
ENF = evergreen needleleaf forest
GRA = grasslands
MF = mixed forest
OSH = open shrublands
SAV = savannah
WET = wetlands

Ecosystem function axes

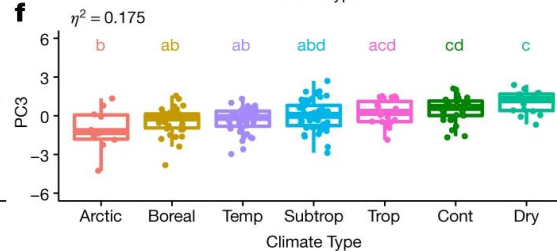
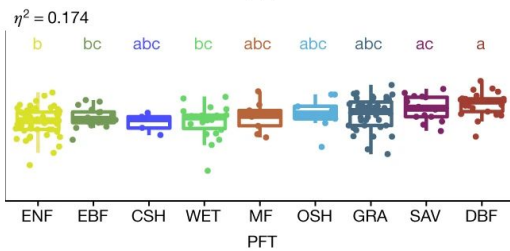
Max productivity



Water use



Carbon use



Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

Published online: 22 September 2021

Open access

Check for updates

Mirco Migliavacca^{1,2,85,12}, Talie Musavi¹, Miguel D. Mahecha^{12,8,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,8}, Dennis D. Baldocchi¹, Oscar Perez-Priego¹, Rune Christiansen¹, Jonas Peters¹, Karen Anderson¹, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹⁰, Damien Bone¹², Nina Buchmann¹, Silvia Caldararu¹, Arnaud Carrara¹, Nuno Carvalhais¹⁰, Alessandro Ceccati¹, Jiquan Chen¹, Jamie Cleverly^{10,11}, Edoardo Cremonesi¹, Ankur R. Desai¹, Tarek S. El-Madany¹, Martha M. Farello¹², Marcos Fernández-Martínez²⁴, Gianluca Filippa¹, Matthias Forkel¹², Marta Galvagno¹, Ulisse Gomarasca¹, Christopher M. Gough¹⁰, Mathias Göckede¹, Andreas Ibrom¹², Hiroki Iwasa¹⁰, Ivan A. Janssens¹, Martin Jung¹, Jens Kattge¹, Trevor F. Keenan^{10,11}, Alexander Knohl^{10,11}, Hideki Kobayashi¹², Guido Krömer^{10,11}, Beverly E. Law¹⁰, Michael J. Liddell¹⁰, Xuanlong Ma¹⁰, Ivan Mammarella¹², David Martin¹, Craig MacFarlane¹⁰, Giorgio Matteucci¹⁰, Leonardo Montagnani^{10,11}, Daniel E. Pabon-Moreno¹, Cinzia Panigada¹, Dario Papale¹², Elise Pendall¹⁴, Josep Penuelas^{10,11}, Richard P. Phillips¹, Peter B. Reich^{14,44,48}, Micol Rossini¹⁰, Eyal Rotenberg¹⁰, Russell L. Scott¹⁰, Clement Stahl¹⁰, Ulrich Weber¹, Georg Wohlfahrt¹⁰, Sebastian Wolf¹⁰, Ian J. Wright^{14,43}, Dan Yakir¹⁰, Sonke Zaehle¹ & Markus Reichstein^{1,2,8,4,12}

CSH = closed shrublands
DBF = deciduous broadleaved forest
DNF = deciduous needleleaf forests
EBF = evergreen broadleaved forest
ENF = evergreen needleleaf forest
GRA = grasslands
MF = mixed forest
OSH = open shrublands
SAV = savannah
WET = wetlands

Ecosystem function axes

What traits best explain this variation?

- Compared plant traits vs climate traits

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

Published online: 22 September 2021

Open access

 Check for updates

Mirco Migliavacca^{1,2,8,9,12}, Talie Musavi¹, Miguel D. Mahecha^{12,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,10}, Dennis D. Baldocchi⁹, Oscar Perez-Priego¹, Rune Christiansen¹, Jonas Peters¹, Karen Anderson¹, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹¹, Damien Bone¹², Nina Buchmann¹, Silvia Caldararu¹, Arnaud Carrara¹, Nuno Carvalhais¹⁰, Alessandro Cecatti¹², Jiquan Chen¹, Jamie Cleverly^{10,11}, Edoardo Cremonesi¹², Ankur R. Desai¹², Tarek S. El-Madany¹, Martha M. Farella¹², Marcos Fernández-Martínez²⁴, Gianluca Filippa¹², Matthias Forkel¹², Marta Galvagno¹², Ulisse Gomarasca¹, Christopher M. Gough¹⁰, Mathias Göckede¹, Andreas Ibrom¹², Hiroki Ikawa¹⁰, Ivan A. Janssens¹⁰, Martin Jung¹, Jens Kattge¹², Trevor F. Keenan^{10,11}, Alexander Knohl^{10,12}, Hideki Kobayashi¹², Guido Kraemer^{10,11}, Beverly E. Law¹⁰, Michael J. Liddell¹², Xuanlong Ma¹⁰, Ivan Mammarella¹², David Martin¹, Craig Macfarlane¹⁰, Giorgio Matteucci¹⁰, Leonardo Montagnani^{10,11}, Daniel E. Pabon-Moreno¹, Cinzia Panigada¹², Dario Papale¹², Elise Pendall¹⁴, Josep Penuelas^{10,11}, Richard P. Phillips¹², Peter B. Reich^{14,10,11}, Micol Rossini¹², Eyal Rotenberg¹⁰, Russell L. Scott¹¹, Clement Stahl¹², Ulrich Weber¹, Georg Wohlfahrt¹⁰, Sebastian Wolf¹⁰, Ian J. Wright^{14,11}, Dan Yaki¹⁰, Sönke Zaehe¹ & Markus Reichstein^{1,2,3,4,12}

Ecosystem function axes

Plant traits:

- Nitrogen content (N%)
- maximum leaf area index (LAI_{max})
- maximum canopy height (H_c)
- above-ground biomass (AGB)

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

Published online: 22 September 2021

Open access

 Check for updates

Mirco Migliavacca^{1,2,8,9,10}, Talie Musavi¹, Miguel D. Mahecha^{12,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{10,6}, Dennis D. Baldocchi⁸, Oscar Perez-Priego¹, Rune Christiansen¹, Jonas Peters⁴, Karen Anderson⁷, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹⁰, Damien Bone¹², Nina Buchmann¹, Silvia Caldararu¹, Arnaud Carrara¹, Nuno Carvalhais¹⁰, Alessandro Cecatti¹², Jiquan Chen¹, Jamie Cleverly^{10,20}, Edoardo Cremonesi¹², Ankur R. Desai¹², Tarek S. El-Madany¹, Martha M. Farella¹², Marcos Fernández-Martínez²⁴, Gianluca Filippa¹², Matthias Forkel¹², Marta Galvagno¹⁰, Ulisse Gomasasca¹, Christopher M. Gough¹⁹, Mathias Göckede¹, Andreas Ibrom¹⁷, Hiroki Iwasa¹⁸, Ivan A. Janssens¹⁴, Martin Jung¹, Jens Kattge¹, Trevor F. Keenan^{1,20}, Alexander Knohl^{10,20}, Hideki Kobayashi¹², Guido Kraemer^{1,19}, Beverly E. Law²¹, Michael J. Liddell¹², Xuanlong Ma¹⁰, Ivan Mammarella¹², David Martin¹, Craig Macfarlane¹⁰, Giorgio Matteucci¹⁹, Leonardo Montagnani^{10,41}, Daniel E. Pabon-Moreno¹, Cinzia Panigada¹², Dario Papale¹², Elise Pendall¹⁴, Josep Penuelas^{10,48}, Richard P. Phillips¹², Peter B. Reich^{14,48,49}, Micol Rossini¹², Eyal Rotenberg¹⁰, Russell L. Scott¹⁴, Clement Stahl¹², Ulrich Weber¹, Georg Wohlfahrt¹⁰, Sebastian Wolf¹⁰, Ian J. Wright^{14,43}, Dan Yaki¹⁰, Sönke Zaehle¹ & Markus Reichstein^{1,2,34,12}

Ecosystem function axes

Climate traits:

- cumulative soil water availability index (CSWI),
- mean annual precipitation (P)
- mean shortwave incoming radiation (SWin)
- mean air temperature (Tair)
- mean vapour pressure deficit (VPD)

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

Published online: 22 September 2021

Open access

 Check for updates

Mirco Migliavacca^{1,2,8,9,10}, Talie Musavi¹, Miguel D. Mahecha^{12,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,10}, Dennis D. Baldocchi⁹, Oscar Perez-Priego¹, Rune Christiansen¹, Jonas Peters¹, Karen Anderson¹, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹⁰, Damien Bone¹², Nina Buchmann¹, Silvia Caldararu¹, Arnaud Carrara¹, Nuno Carvalhais¹⁰, Alessandro Cecatti¹², Jiquan Chen¹, Jamie Cleverly^{10,11}, Edoardo Cremonesi¹², Ankur R. Desai¹², Tarek S. El-Madany¹, Martha M. Farella¹², Marcos Fernández-Martínez²⁴, Gianluca Filippa¹², Matthias Forkel¹², Marta Galvagno¹², Ulisse Gomarasca¹, Christopher M. Gough¹⁰, Mathias Göckede¹, Andreas Ibrom¹², Hiroki Ikawa¹⁰, Ivan A. Janssens¹⁰, Martin Jung¹, Jens Kattge¹, Trevor F. Keenan^{1,10}, Alexander Knohl^{10,12}, Hideki Kobayashi¹², Guido Kraemer^{1,10}, Beverly E. Law¹⁰, Michael J. Liddell¹², Xuanlong Ma¹⁰, Ivan Mammarella¹², David Martin¹, Craig MacFarlane¹⁰, Giorgio Matteucci¹⁰, Leonardo Montagnani^{10,11}, Daniel E. Pabon-Moreno¹, Cinzia Panigada¹², Dario Papale¹², Elise Pendall¹⁴, Josep Penuelas^{10,12}, Richard P. Phillips¹², Peter B. Reich^{14,10,12}, Micol Rossini¹², Eyal Rotenberg¹⁰, Russell L. Scott¹¹, Clement Stahl¹², Ulrich Weber¹, Georg Wohlfahrt¹⁰, Sebastian Wolf¹⁰, Ian J. Wright^{14,12}, Dan Yaki¹⁰, Sönke Zaehe¹ & Markus Reichstein^{1,2,3,4,12}

Ecosystem function axes

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

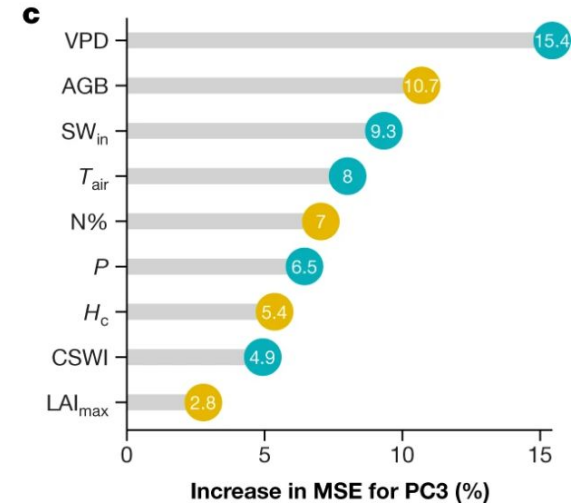
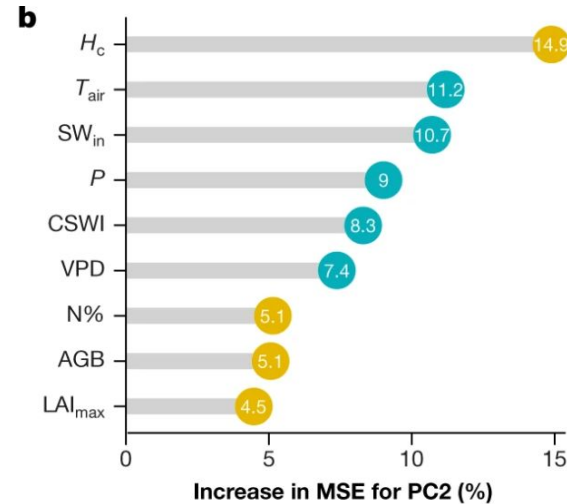
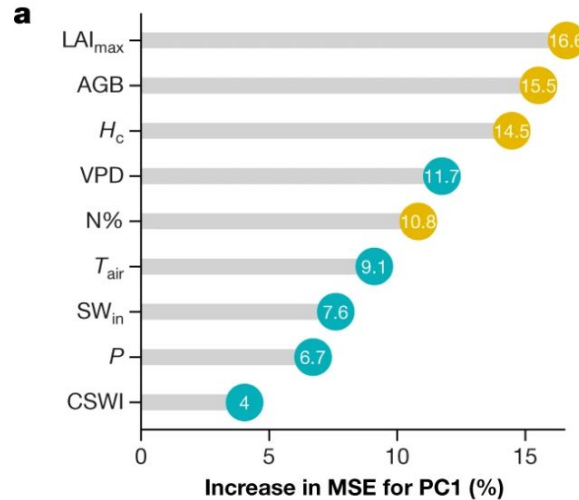
Accepted: 20 August 2021

Published online: 22 September 2021

Open access

 Check for updates

Mirco Migliavacca^{1,2,85,†}, Tatle Musavi¹, Miguel D. Mahecha^{12,8,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,86}, Dennis D. Baldocchi¹, Oscar Perez-Priego¹, Rune Christiansen¹, Jonas Peters¹, Karen Anderson¹, Michael Bahn¹⁰, T. Andrew Black¹¹, Peter D. Blanken¹², Damien Bonal¹³, Nina Buchmann¹⁴, Silvia Calderaru¹⁵, Arnaud Carrara¹⁶, Nuno Carvalhais¹⁰, Alessandro Cecatti¹⁷, Jiquan Chen¹⁸, Jamie Cleverly^{19,20}, Edoardo Cremonesi²¹, Ankur R. Desai²², Tarek S. El-Madany²³, Martha M. Farella²⁴, Marcos Fernández-Martínez²⁴, Gianluca Filippa²⁵, Matthias Forkel²⁶, Marta Galvagno²⁷, Ulisse Gomasasca²⁸, Christopher M. Gough²⁹, Mathias Göckede³⁰, Andreas Ibrom³¹, Hiroki Iwasa³², Ivan A. Janssens³³, Martin Jung³⁴, Jens Kattge³⁵, Trevor F. Keenan^{36,37}, Alexander Knohl^{38,39}, Hideki Kobayashi⁴⁰, Guido Krömer^{41,42}, Beverly E. Law⁴³, Michael J. Liddell⁴⁴, Xuanlong Ma⁴⁵, Ivan Mammarella⁴⁶, David Martin⁴⁷, Craig MacFarlane⁴⁸, Giorgio Matteucci⁴⁹, Leonardo Montagnani^{50,51}, Daniel E. Pabon-Moreno⁵², Cinzia Panigada⁵³, Dario Papale⁵⁴, Elise Pendall⁵⁵, Josep Penuelas^{56,57}, Richard P. Phillips⁵⁸, Peter B. Reich^{59,60,61}, Micol Rossini⁶², Eyal Rotenberg⁶³, Russell L. Scott⁶⁴, Clement Stah⁶⁵, Ulrich Weber⁶⁶, Georg Wohlfahrt⁶⁷, Sebastian Wolf⁶⁸, Ian J. Wright^{69,70}, Dan Yakir⁷¹, Sönke Zaehe⁷² & Markus Reichstein^{1,2,84,†}



PC1 = maximum ecosystem productivity PC2 = ecosystem water-use strategies PC3 = ecosystem carbon-use efficiency

Ecosystem function axes

Note:

These types of studies are always sensitive to which traits, which services, etc.

Still, shows how important traits can be for linking with function

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

Accepted: 20 August 2021

Published online: 22 September 2021

Open access

 Check for updates

Mirco Migliavacca^{1,2,85,†}, Talie Musavi¹, Miguel D. Mahecha^{1,2,3,4}, Jacob A. Nelson¹, Jürgen Knauer^{1,16}, Dennis D. Baldocchi⁸, Oscar Perez-Priego¹, Rune Christiansen¹, Jonas Peters¹, Karen Anderson¹, Michael Bahn¹⁶, T. Andrew Black¹, Peter D. Blanken¹, Damien Bone¹⁷, Nina Buchmann¹, Silvia Caldararu¹, Arnaud Carrara¹, Nuno Carvalhais¹⁸, Alessandro Cecatti¹⁷, Jiquan Chen¹, Jamie Cleverly^{19,20}, Edoardo Cremonesi²¹, Ankur R. Desai²², Tarek S. El-Madany¹, Martha M. Farella²³, Marcos Fernández-Martínez²⁴, Gianluca Filippa²⁵, Matthias Forkel²⁶, Marta Galvagno²⁷, Ulisse Gomasasca¹, Christopher M. Gough²⁸, Mathias Göckede¹, Andreas Ibrom²⁷, Hiroki Iikawa²⁹, Ivan A. Janssens³⁰, Martin Jung¹, Jens Kattge¹, Trevor F. Keenan^{31,32}, Alexander Knohl^{33,34}, Hideki Kobayashi³⁵, Guido Kraemer^{3,19}, Beverly E. Law³⁶, Michael J. Liddell³⁷, Xuanlong Ma³⁸, Ivan Mammarella³⁹, David Martin¹, Craig Macfarlane³⁸, Giorgio Matteucci³⁹, Leonardo Montagnani^{40,41}, Daniel E. Pabon-Moreno¹, Cinzia Panigada⁴², Dario Papale⁴³, Elise Pendall⁴⁴, Josep Penuelas^{45,46}, Richard P. Phillips⁴⁷, Peter B. Reich^{44,46,48}, Micol Rossini⁴⁹, Eyal Rotenberg⁵⁰, Russell L. Scott⁵¹, Clement Stahl⁵², Ulrich Weber¹, Georg Wohlfahrt⁵³, Sebastian Wolf⁵⁴, Ian J. Wright^{44,55}, Dan Yakir⁵⁶, Sönke Zaehle¹ & Markus Reichstein^{1,2,34,57}

Which organisms are most important?

Mass ratio hypothesis:

- The traits of the species that account for most of the biomass drive functioning
- Grime noted that that this only refers to autotrophs

*Journal of
Ecology* 1998,
86, 902–910

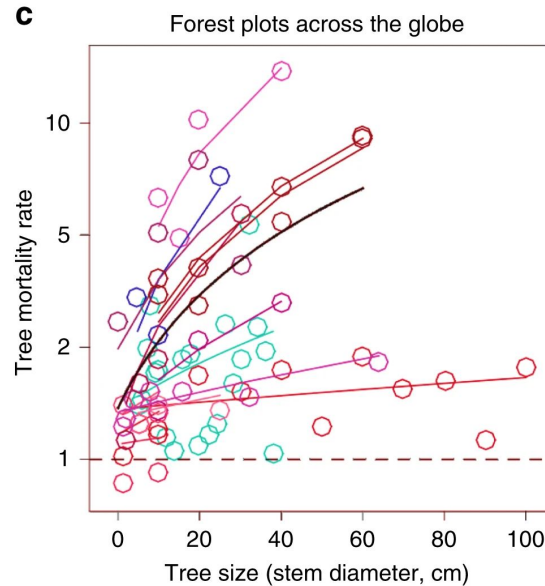
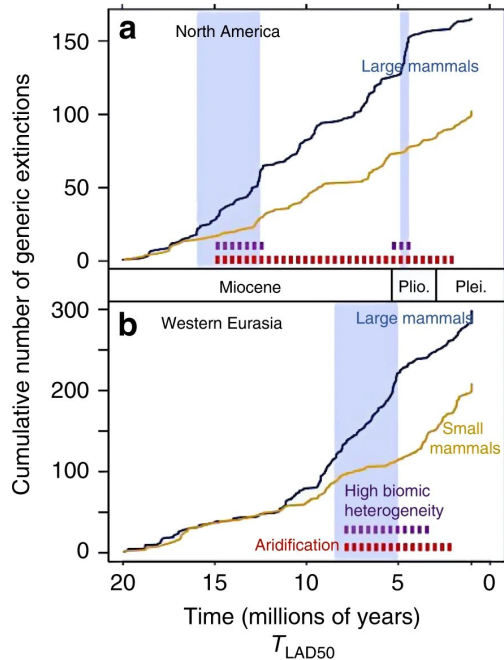
ESSAY REVIEW

Benefits of plant diversity to ecosystems: immediate, filter and founder effects

J.P. GRIME

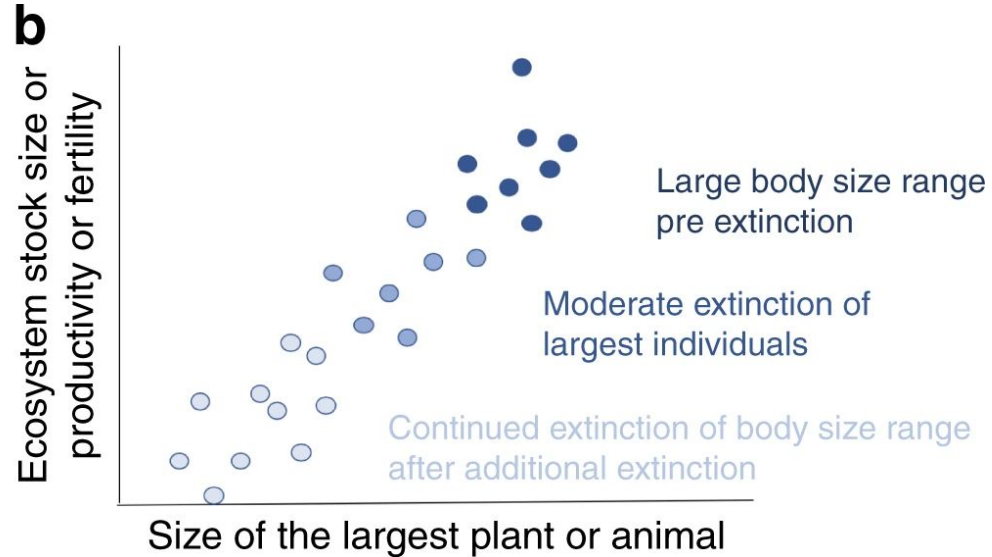
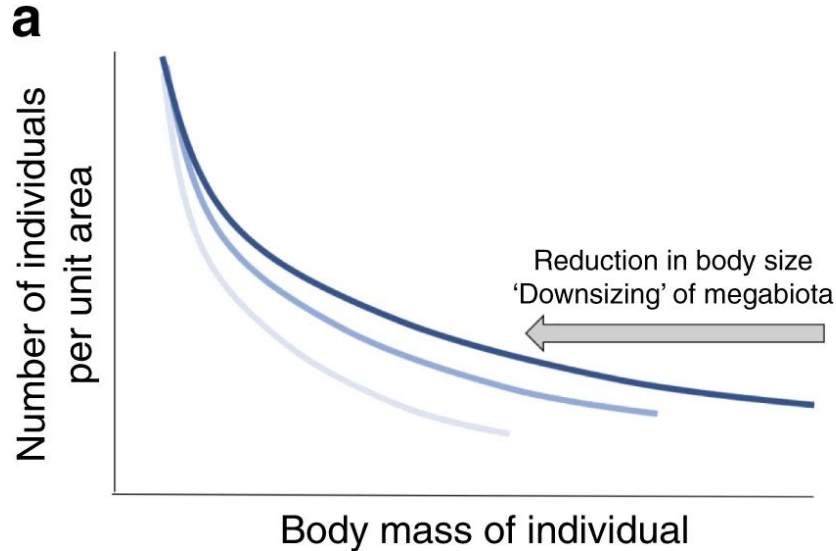
Which organisms are most important?

The big things drive functions



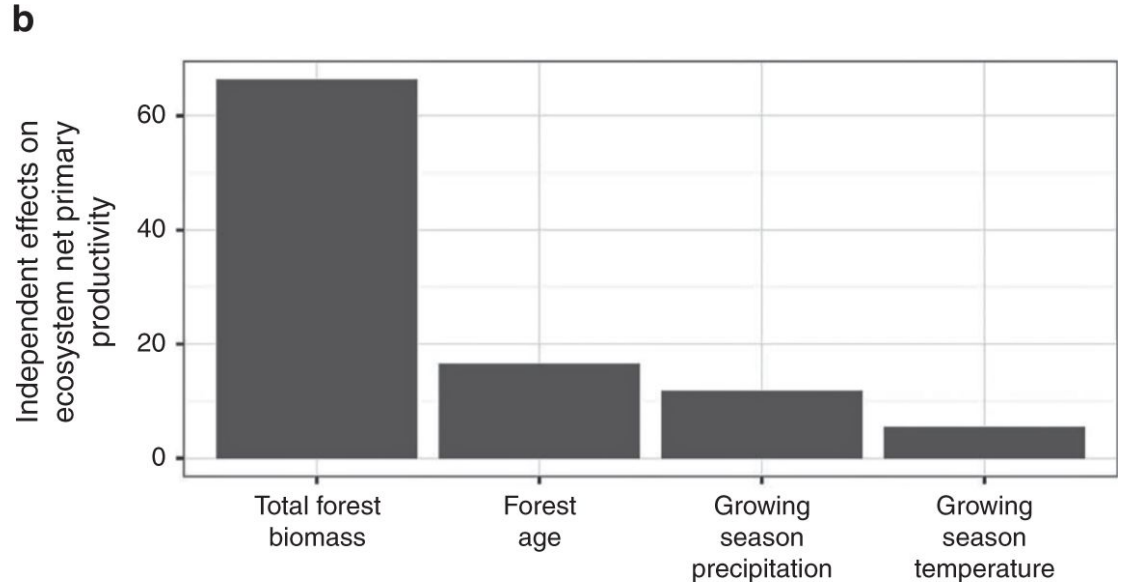
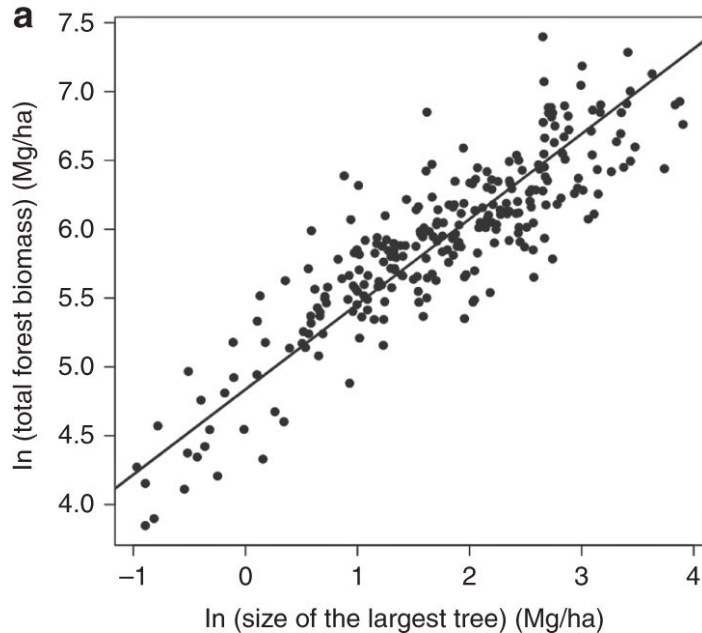
Which organisms are most important?

The big things drive functions



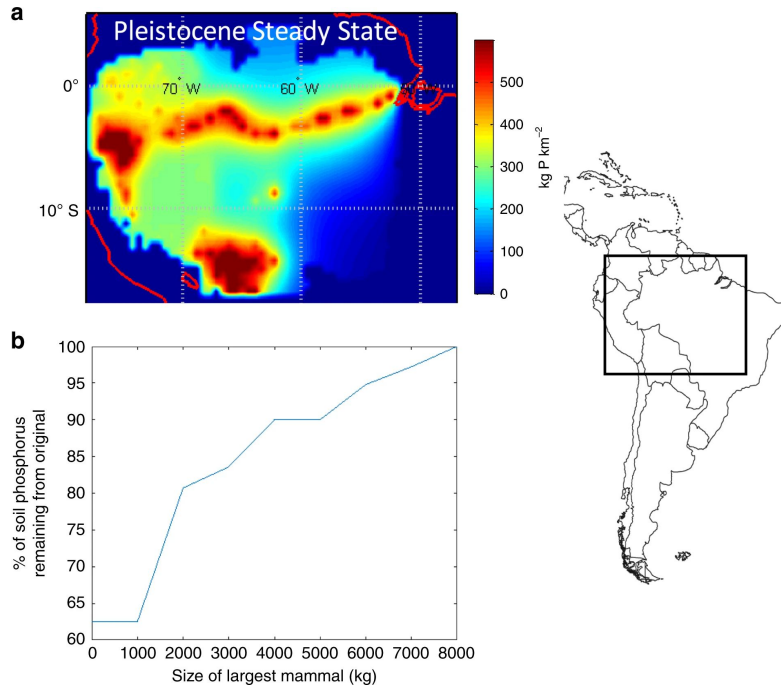
Which organisms are most important?

The big things drive functions

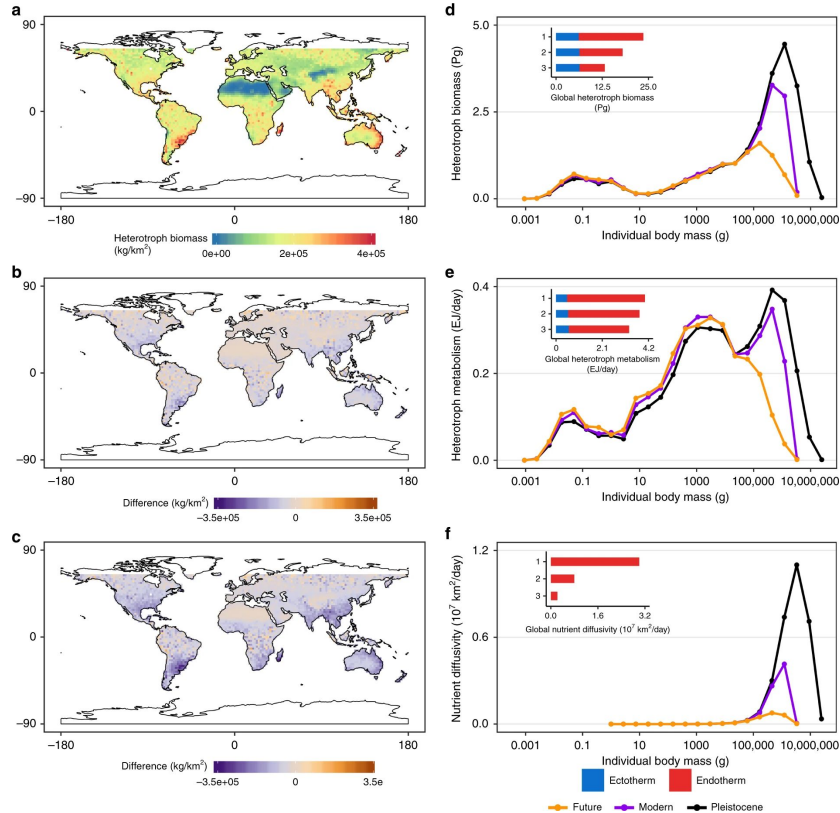


Which organisms are most important?

The big things drive functions



Which organisms are most important?



Megafauna services

Q: How has megafauna loss impacted nutrients?

Megafauna services

Q: How has megafauna loss impacted nutrients?

Approach: models based on traits

Traits used:

- Body size (M)
- Range size (DD)
- Metabolic rate (MR)
- Population density (PD)
- Food passage time (PR)
- Total plant biomass (B)
- Edible fraction of plant biomass (α)

Megafauna services

Q: How has megafauna loss impacted nutrients?

Approach: models based on traits

Traits used:

- Body size (M)
- Range size (DD)
- Metabolic rate (MR)
- Population density (PD)
- Food passage time (PR)
- Total plant biomass (B)
- Edible fraction of plant biomass (α)

$$\Phi = (1 - \epsilon) * MR * \frac{PD}{\alpha B} * \frac{(DD * PR)^2}{2 * PR} = \frac{0.78 * 0.05 * M^{1.17}}{\alpha B} \quad (1)$$

Megafauna services

Q: How has megafauna loss impacted nutrients?

Approach: models based on traits

Traits used:

- Body size (M)
- Range size (DD)
- Metabolic rate (MR)
- Population density (PD)
- Food passage time (PR)
- Total plant biomass (B)
- Edible fraction of plant biomass (α)

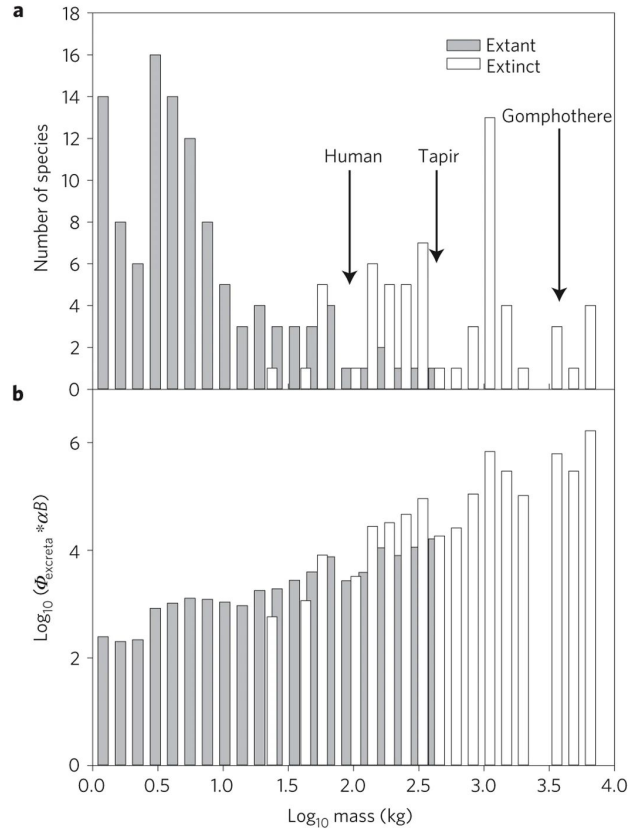
$$\Phi = (1 - \epsilon) * MR * \frac{PD}{\alpha B} * \frac{(DD * PR)^2}{2 * PR} = \frac{0.78 * 0.05 * M^{1.17}}{\alpha B} \quad (1)$$

Note: superlinear scaling means bigger things are disproportionately important!

The legacy of the Pleistocene megafauna extinctions on nutrient availability in Amazonia

Christopher E. Doughty¹*, Adam Wolf² and Yadvinder Malhi¹

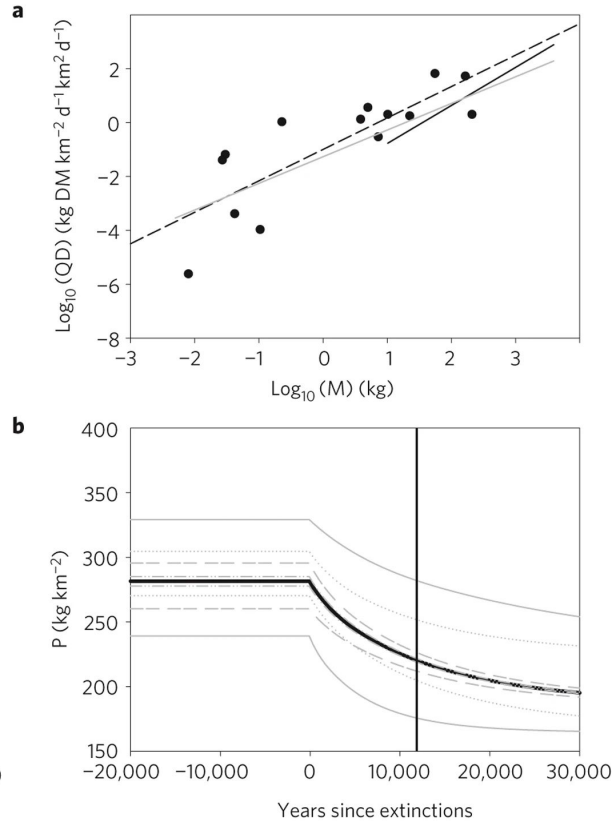
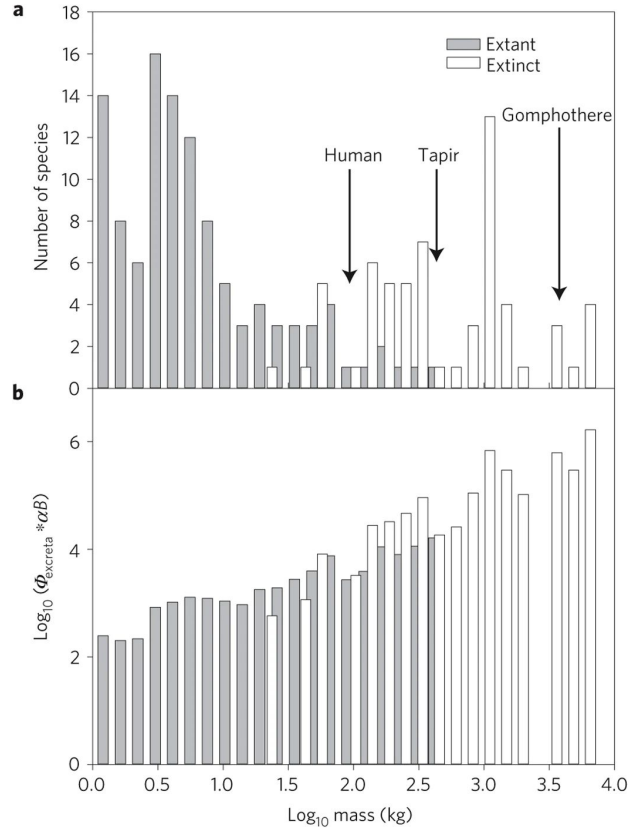
Megafauna services



The legacy of the Pleistocene megafauna extinctions on nutrient availability in Amazonia

Christopher E. Doughty^{1*}, Adam Wolf² and Yadvinder Malhi¹

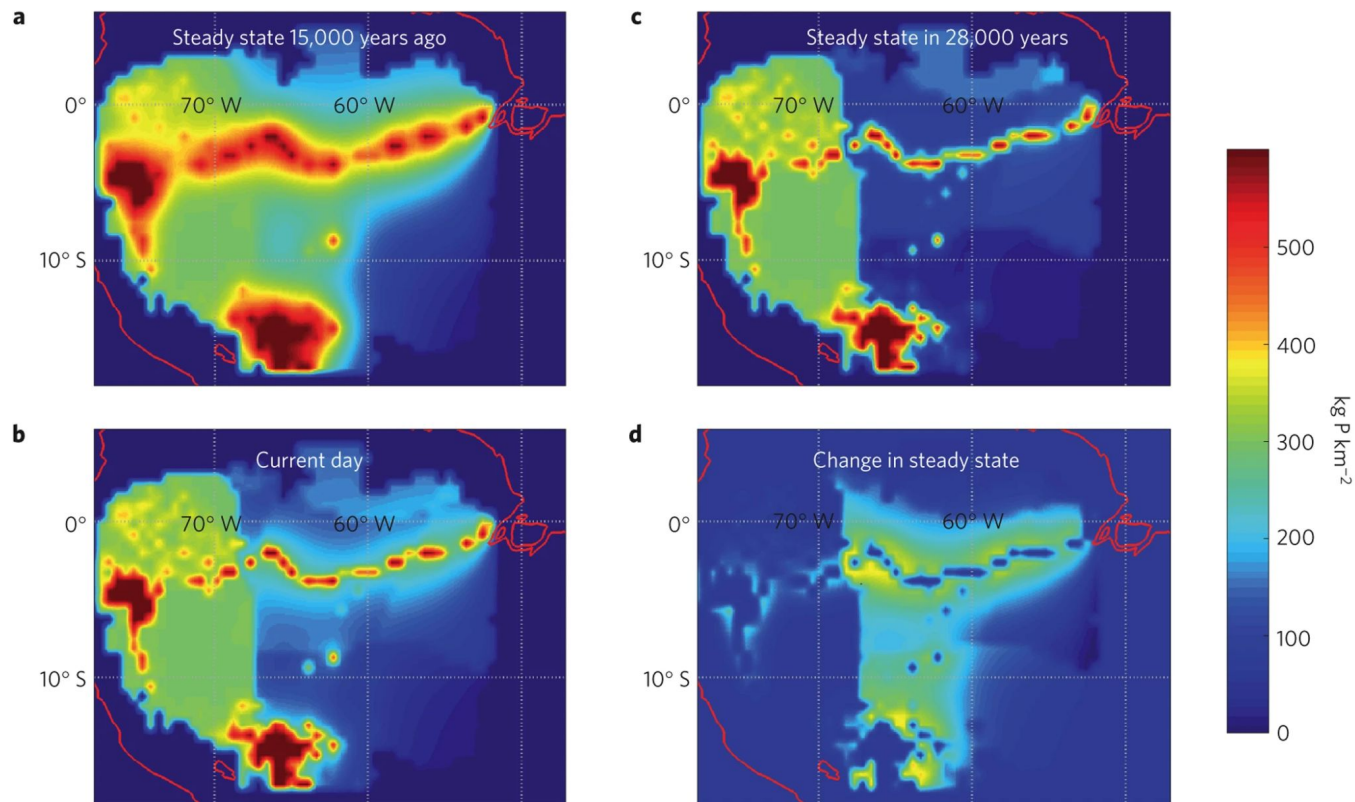
Megafauna services



Megafauna services

The legacy of the Pleistocene megafauna extinctions on nutrient availability in Amazonia

Christopher E. Doughty¹*, Adam Wolf² and Yadvinder Malhi¹



The legacy of the Pleistocene megafauna
extinctions on nutrient availability in Amazonia

Christopher E. Doughty¹*, Adam Wolf² and Yadvinder Malhi¹

Megafauna services

Table 1 | Average $\Phi_{\text{excreta}} * \alpha B$ (km² yr⁻¹) for each continent calculated for modern species and modern plus extinct species.

	North America	South America	Australia	Eurasia	Africa
Number of species extinct	65	64	45	9	13
Mean weight of extinct animals (kg)	846	1,156	188	2,430	970
Modern $\Phi_{\text{excreta}} * \alpha B$	13,876	12,934	21,804	21,779	265,621
Modern + extinct fauna $\Phi_{\text{excreta}} * \alpha B$	140,716 ($\pm 38,000$)	283,854 ($\pm 81,000$)	48,250 ($\pm 8,000$)	118,349 ($\pm 29,000$)	324,848 ($\pm 18,000$)
Percentage of original	10% ($\pm 2\%$)	5% ($\pm 1\%$)	45% ($\pm 6\%$)	18% ($\pm 4\%$)	82% ($\pm 4\%$)

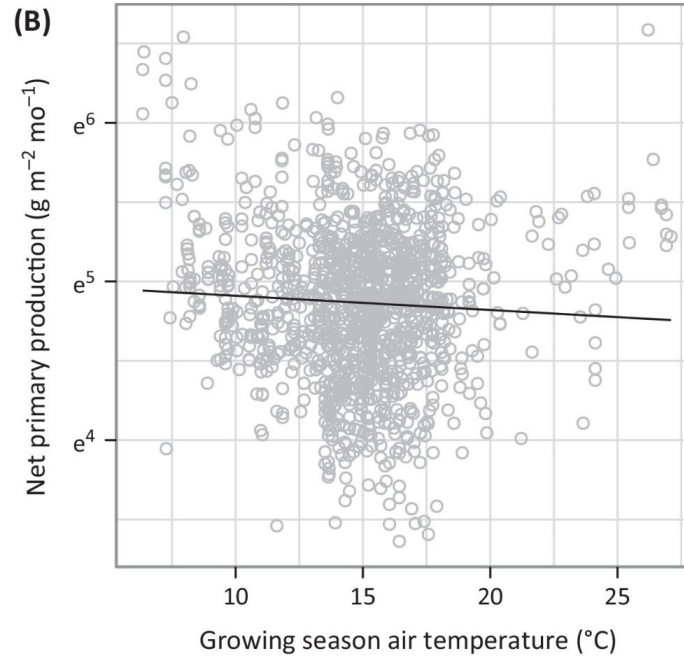
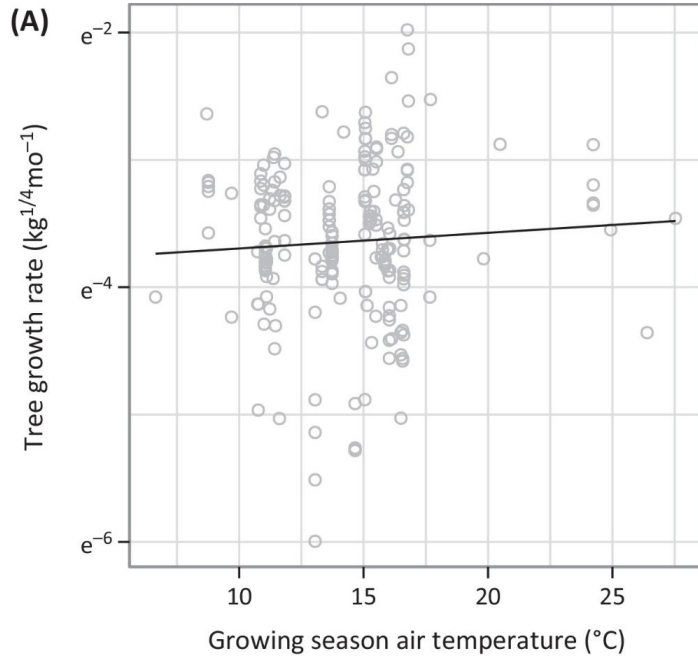
Bottom row is the percentage of the original $\Phi_{\text{excreta}} * \alpha B$ remaining. The error represents an uncertainty in extinct species distribution of 30%.

Opinion

Plant Thermoregulation:
Energetics, Trait–Environment
Interactions, and Carbon
Economics

Sean T. Michaletz,^{1,2,*} Michael D. Weiser,³ Jizhong Zhou,^{4,5,6}
Michael Kasprow,^{7,8} Brent R. Helliker,⁹ and
Brian J. Enquist^{1,9,10,11}

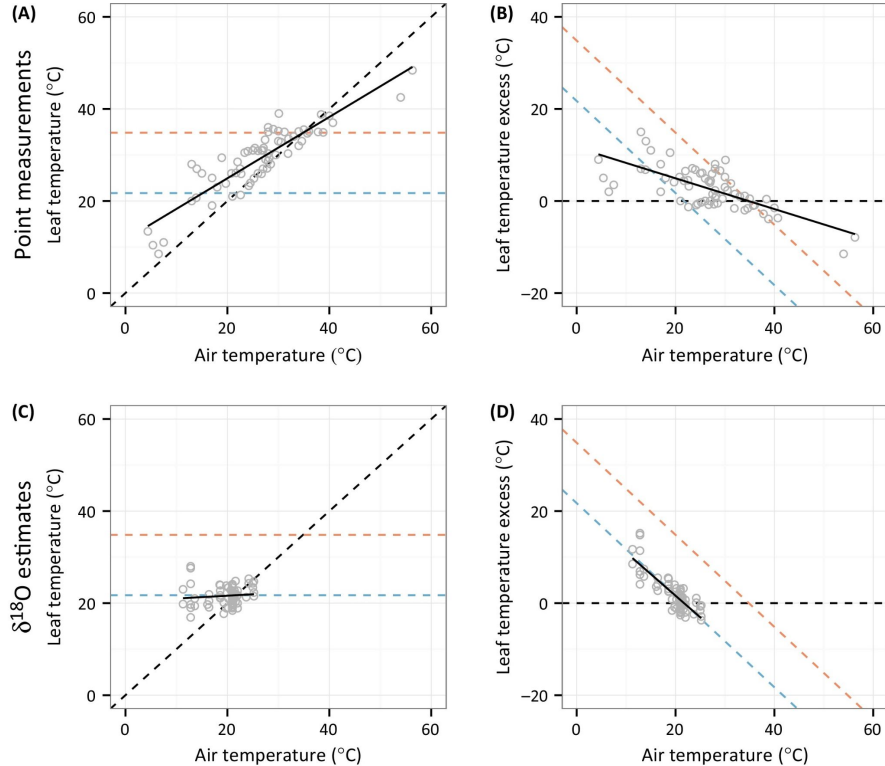
Traits and temp and photosynthesis



Trends in Ecology & Evolution

Photosynthesis is temp-dependent, but not much global signal of temperature

Traits and temp and photosynthesis



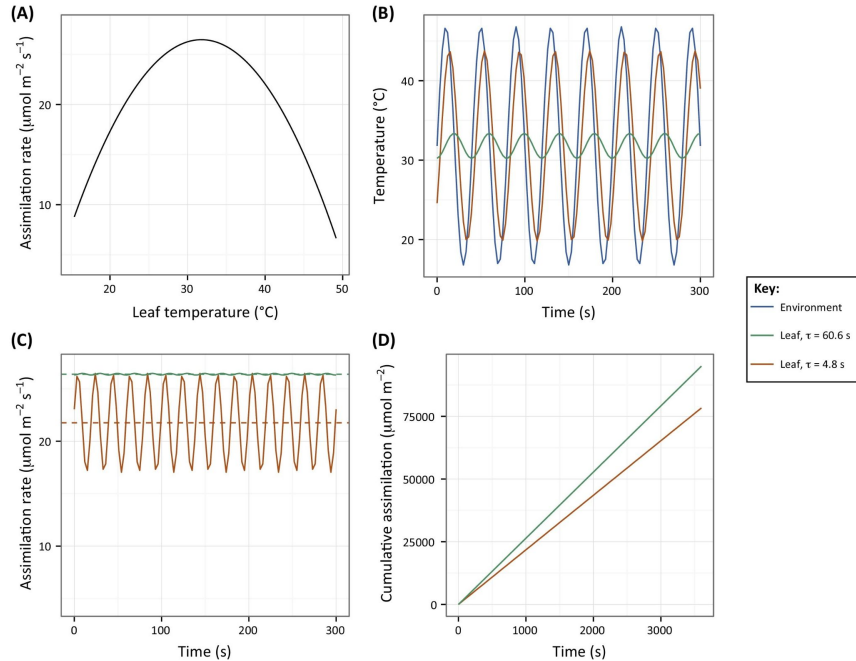
Opinion

Plant Thermoregulation: Energetics, Trait–Environment Interactions, and Carbon Economics

Sean T. Michaletz,^{1,2,*} Michael D. Weiser,³ Jizhong Zhou,^{4,5,6}
Michael Kaspari,^{7,8} Brent R. Helliker,⁹ and
Brian J. Enquist^{1,9,10,11}

Traits and temp and photosynthesis

If we assume two leaves sharing a common photosynthesis temperature response (A) and environment (B), but different thermal time constants τ (B)



Then energy budgets predict differences in time-averaged assimilation rates (C) and time-integrated carbon assimilation (D).

Opinion

Plant Thermoregulation: Energetics, Trait–Environment Interactions, and Carbon Economics

Sean T. Michaletz,^{1,2,*} Michael D. Weiser,³ Jizhong Zhou,^{4,5,6} Michael Kaspari,^{5,7} Brent R. Helliker,⁸ and Brian J. Enquist^{1,9,10,11}

Traits and temp and photosynthesis

We need to predict photosynthetic rate

Photosynthetic rate depends on temperature

Leaf temperature can't be predicted from air temp

How do we get at leaf temp?

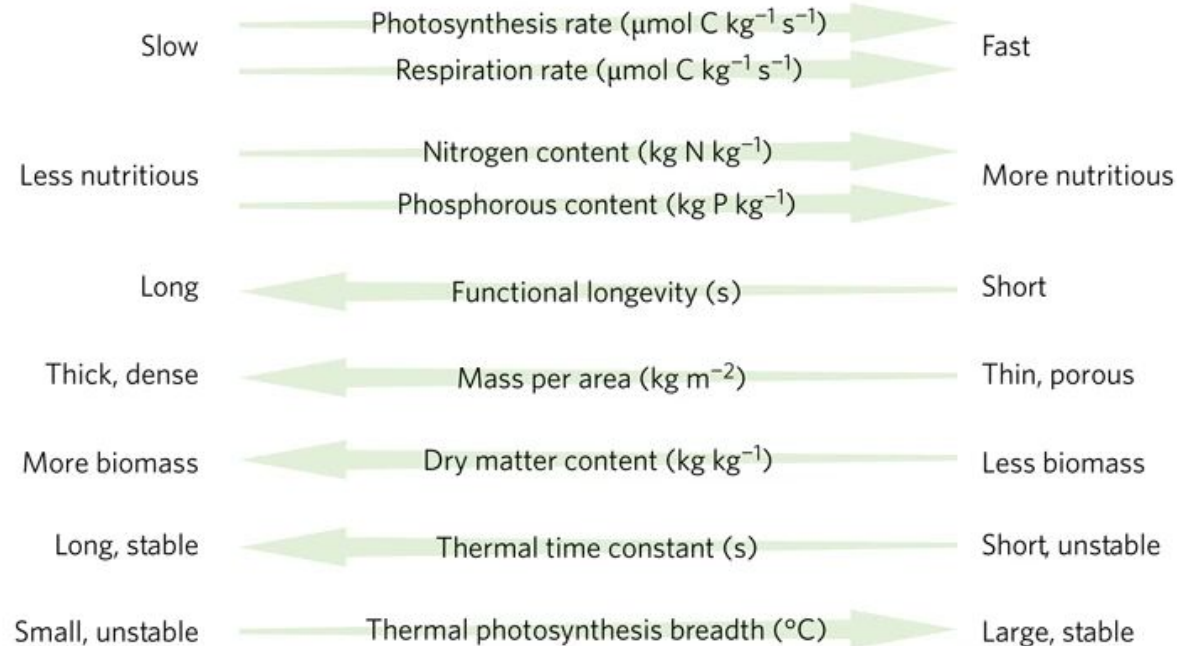
The energetic and carbon economic origins of leaf thermoregulation

Sean T. Michaletz^{1,2*}, Michael D. Weiser¹, Nate G. McDowell¹, Jizhong Zhou^{4,5,6}, Michael Kaspar^{3,7}, Brent R. Helliker⁸ and Brian J. Enquist^{2,9,10,11}

The energetic and carbon economic origins of leaf thermoregulation

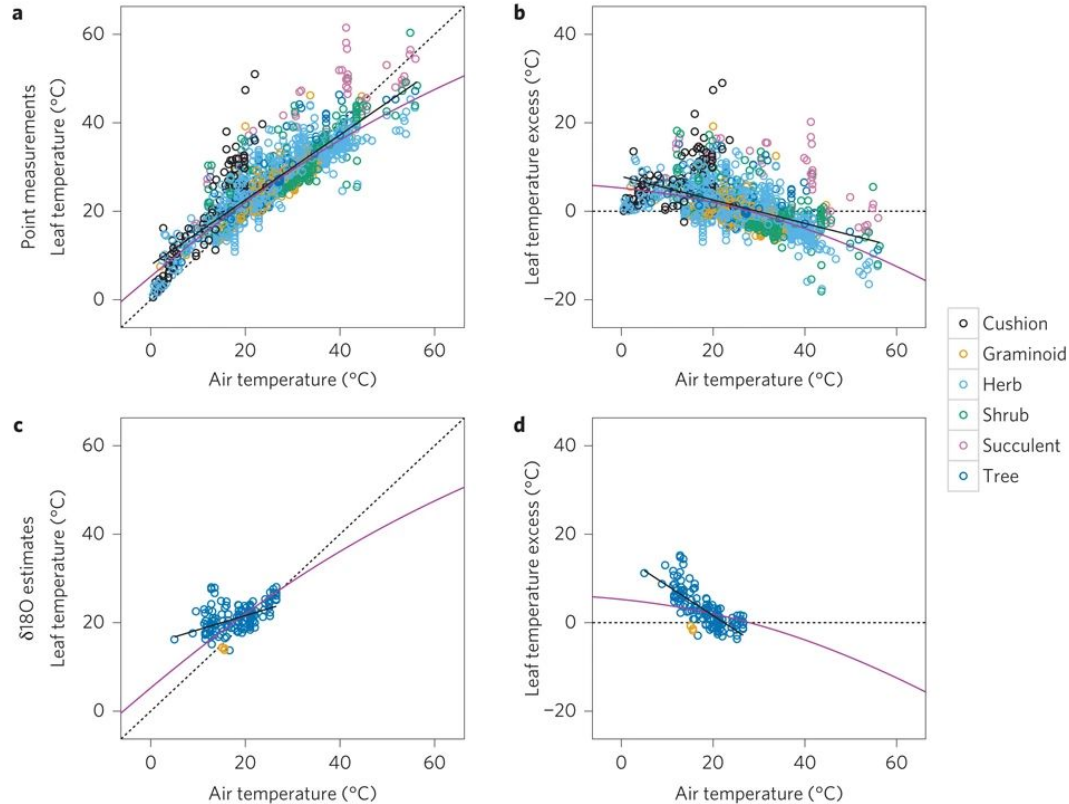
Sean T. Michaletz^{1,2*}, Michael D. Weiser¹, Nate G. McDowell¹, Jizhong Zhou^{4,5,6}, Michael Kaspar^{1,7}, Brent R. Helliker⁸ and Brian J. Enquist^{2,9,10,11}

Traits and temp and photosynthesis



The energetic and carbon economic origins of leaf thermoregulation

Sean T. Michaletz^{1,2*}, Michael D. Weiser¹, Nate G. McDowell¹, Jizhong Zhou^{4,5,6}, Michael Kaspar^{3,7}, Brent R. Helliker⁸ and Brian J. Enquist^{2,9,10,11}



Pink = prediction from trait-based model

Black = best fit regression

Traits and temp and photosynthesis

Studies and theory like this represent major advances
in linking form and function

The energetic and carbon economic origins of leaf thermoregulation

Sean T. Michaletz^{1,2*}, Michael D. Weiser¹, Nate G. McDowell¹, Jizhong Zhou^{4,5,6}, Michael Kaspar^{1,7}, Brent R. Helliker⁸ and Brian J. Enquist^{2,9,10,11}

Intraspecific variation matters

Many studies use species level mean trait values

- Turns out, this is a bad idea for estimating functioning

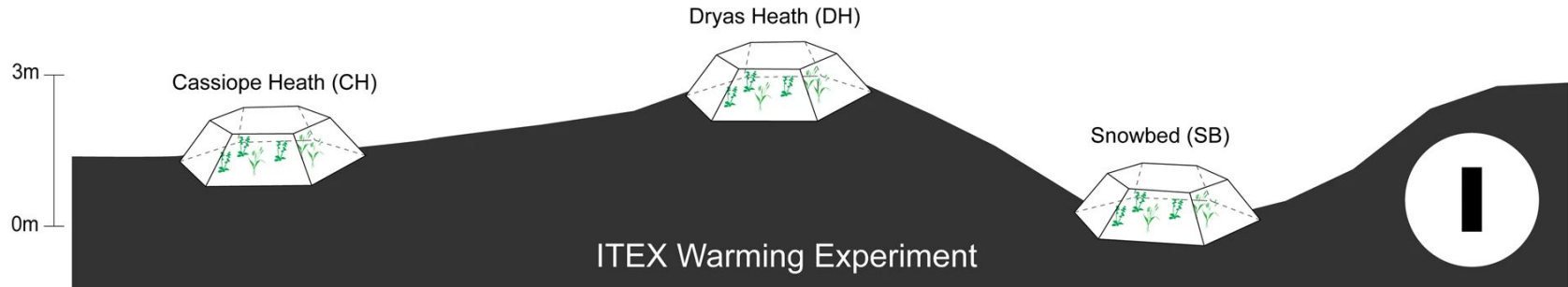
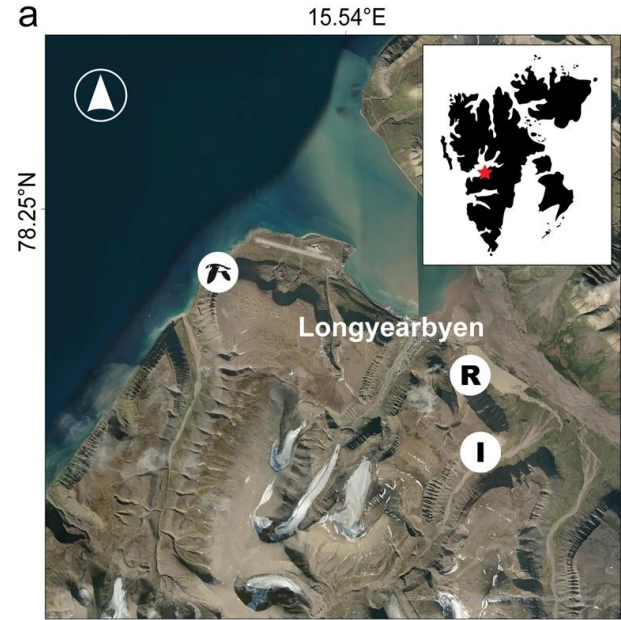
Intraspecific variation matters

Overview:

Measured CO₂ fluxes

Measured traits (including intraspecific variation)

Measured environment



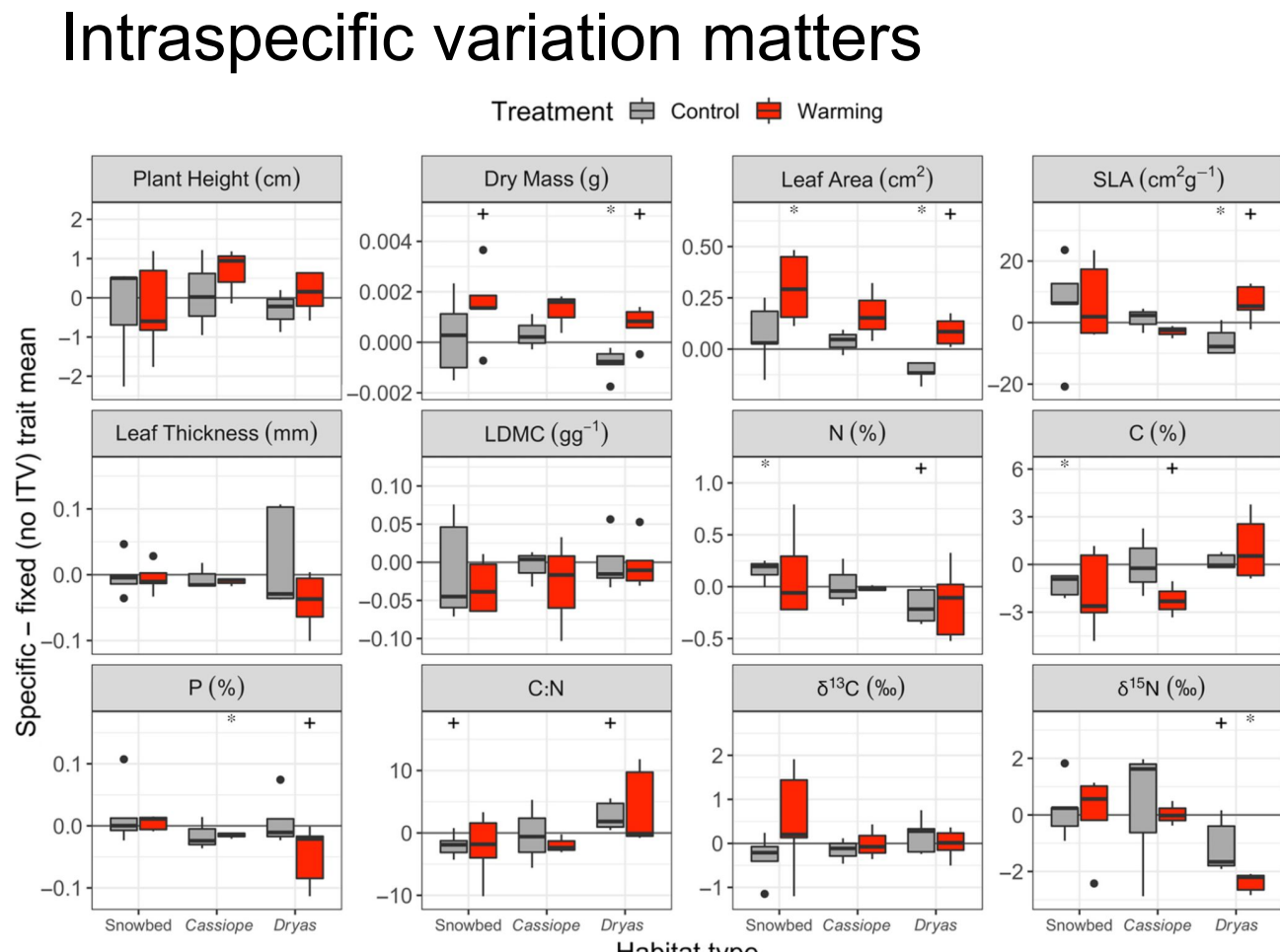
Intraspecific variation matters

To estimate the magnitude of ITV:

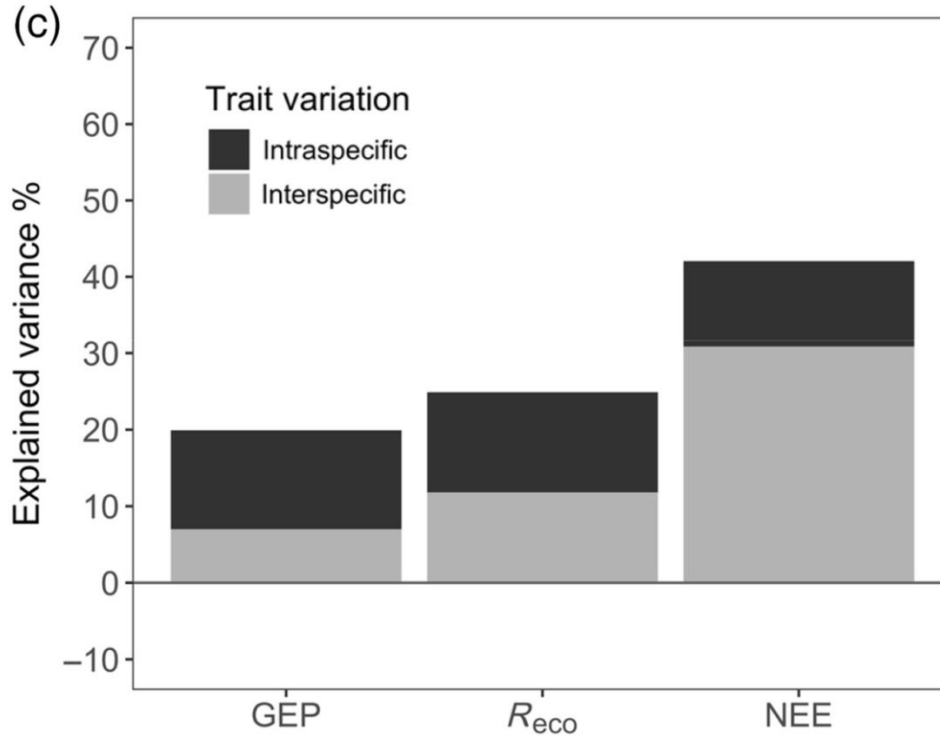
- Compare trait estimates using species mean vs those with ITV

Intraspecific trait variability is a key feature underlying high Arctic plant community resistance to climate warming

Ingibjörg S. Jónsdóttir^{1,2} | Aud H. Halbritter³ | Casper T. Christiansen^{4,5,6} | Inge H. J. Althuisen⁴ | Siri V. Haugum³ | Jonathan J. Henn^{7,8} | Katrin Björnsdóttir⁹ | Brian Salvin Maitner¹⁰ | Yadvinder Malhi¹¹ | Sean T. Michaletz¹² | Ruben E. Roos¹³ | Kari Klenderud¹³ | Hanna Lee^{5,14} | Brian J. Enquist¹⁵ | Vigdis Vandvik^{3,4}



Intraspecific variation matters



GEP = gross ecosystem productivity

R_{eco} = ecosystem respiration

NEE = net ecosystem CO₂ exchange

$GEP = NEE + R_{eco}$

Takeaways

- Traits help us understand variation in ecosystem services
- Traits - environment interactions can be important
- Mechanistic trait models can provide important insights
- Intraspecific trait variation is super important
- Some species are more important than others

Next class: Traits and global change